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Volume 01 — M00–M11

Information Security (CEC) · SWAYAM / CEC Information Security **notes-3** — concept exam notes (definitions & what things do). Breadth over lab depth.

M00: Introductory Video — Information Security

Learning focus

- State why information security now spans individuals and enterprises.
- List the course's promised themes: web trust, DNS, HTTPS, client-side attacks, and secure coding.
- Explain the coordinator's claim that poor programming is the root cause of most information-security threats.
- Recall that this is a short, two-credit course on information and network security.

Key terms

Term	Meaning in this lecture
Security loophole	Organizational gap that lets an attacker break in and exploit a vulnerability
Authoritative DNS reply	Name-server answer that should be trusted when judging whether a site is genuine
HTTPS digital signature	Cryptographic check that the web server is the claimed site
DNS poisoning	Tampering with name resolution so users reach a fake host
Client-side attack	Compromise that succeeds because the user (browser) approaches a malicious site
Secure coding	The course's stated end-goal: stop shipping the vulnerabilities that attacks exploit

Exam lines

- Security is ambient: individual and enterprise alike.
- Trust on the web is technical (DNS + HTTPS), not visual.
- Poor programming, especially copy-paste reuse, is framed as the root cause of threats.
- Client-side attacks are hard to stop once the user cooperates with the attacker.
- The two-credit course aims at network, OS, database, DNS, and secure-coding competence.

M01: Information Security

Learning focus

- Define ethics and contrast consequentialism with deontology as approaches to right and wrong.
- Distinguish system security from information (data) security and state the CIA triad.
- Relate computer-security breaches and overly protective measures to rights, harms, and interests.
- Contrast hacking vs cracking, cyber trespass vs cyber vandalism, and cyber terrorism vs hacktivism vs information warfare.
- Explain why codes of ethics are not enough for information-security professionals.
- State why privacy matters, how IT enables dataveillance, and what concerns arise from public-space tracking and biometrics.

Key terms

Term	Meaning in this lecture
Ethics	Study of right/wrong and good/bad; analyzes behaviours, policies, laws, and social structures
Consequentialism	Wrongness of actions judged by bad consequences
Deontology	Moral duties exist independently of consequences
Computer ethics	1980s field on professional/user duties and IT public-policy ethics
System security	Protecting hardware/software against malicious sabotage
Information security	Protecting stored and transmitted data (CIA)
Safety-critical system	Real-time control system whose failure can threaten life
Hacking / cracking	Unauthorized access; crackers = malicious/disruptive break-ins (hacker self-description)
Hacker ethic	Information should be free; unlimited computer access; cyberspace cannot harm the real world
Cyber trespass	Unauthorized access to systems or password-protected sites
Cyber vandalism	Programs that disrupt networks or corrupt data

Term	Meaning in this lecture
Cyber terrorism	Politically motivated hacking intended to cause grave harm (death and/or severe economic loss)
Hacktivism	Political site disruption without intent to cause serious damage
Information warfare	Using information and attacks on information systems as tools of war
ISSA	Information Systems Security Association; code requires law plus "highest ethical principles" without specifying them
Right to be let alone	Warren and Brandeis's definition of privacy
Informed consent	Tell people how personal data will be stored/used/exchanged and obtain consent
Dataveillance	Large-scale computerized collection/processing of personal data to monitor actions and communications
Biometrics	Identity verification from physiological or behavioral characteristics

Exam lines

- Ethics is not the same as law: legality does not replace morality for individuals or corporations.
- Computer security has a moral structure: CIA plus system vs data, tied to rights, harms, and interests — including economic, cultural, psychological, and life-and-death stakes in safety-critical systems.
- Over-strong or discriminatory security can itself violate access, fairness, and freedom rights (including FOI and free speech when availability is blocked).
- The hacker ethic's "information should be free" claim collides with intellectual property, privacy, and data integrity.
- Cyber terrorism is marked by **political motive plus grave harm**; hacktivism and ordinary cyber crime are not the same thing; information warfare extends ordinary war onto information systems.
- Professional codes (e.g. ISSA) are too thin: security practitioners need ethics **training**, not only a code.
- Privacy is the right to be let alone / control access to private affairs; it is valuable for autonomy and democracy but not absolute.
- IT scaled surveillance into **dataveillance**; 9/11 and consumer tracking intensified the privacy–surveillance tension; public-space tracking and biometrics further squeeze anonymity.

M02: Network Security

Learning focus

- State why e-business networks need network security and list the CIA goals.
- Distinguish a **threat** from an **attack**, and **passive** from **active** attacks.
- Relate snooping, traffic analysis, replay, modification, and denial of service to confidentiality, integrity, and availability.

- Describe cryptography's three mechanisms: symmetric-key, asymmetric-key, and hashing (including MD5 and SHA).
- Place border routers, firewalls (static / stateful / proxy), IDS, and IPS in a network-boundary defense.
- List hallmarks of a good security policy and the non-technical side of defense in depth.

Key terms

Term	Meaning in this lecture
Confidentiality	Protect stored and transmitted information from unauthorized disclosure
Integrity	Authorized entities change data only through authorized mechanisms (faults can also violate it)
Availability	Authorized entities can access information when needed
Threat	Possible danger that might exploit a vulnerability
Attack	Deliberate assault that tries to evade services and violate policy
Passive attack	Eavesdropping / monitoring to obtain transmitted information
Traffic analysis	Inferring communication nature from patterns despite encryption
Snooping	Unauthorized access to or interception of data
Replay	Capture a data unit and retransmit it for an unauthorized effect
Symmetric-key crypto	Same shared secret key for encrypt and decrypt
Asymmetric-key crypto	Related public/private pair; decryption key infeasible to derive from encryption key
RSA	Example algorithm where either related key may encrypt and the other decrypt
Compression function	Fixed-size hash step: n bits in, m bits out, n > m
MD5	Rivest message digest: 512-bit blocks → 128-bit digest (too small)
SHA / NIST	Secure Hash Algorithm family standardized by NIST
ACL	Access control list on a border router (static L3 filter)
Ingress / egress filtering	Only properly destined traffic enters / leaves
Smurf attack	Spoofed ICMP echo to broadcast addresses; responses flood the victim
Stateful firewall	Tracks connections in a state table; most common type
Proxy firewall	Intermediary; full-packet protocol compliance; most advanced, least common
IDS	Detect and alert (signatures / anomaly); does not auto-block
IPS	Detect and automatically prevent/block
Defense in depth	Layered technical controls plus policy and awareness

Exam lines

- E-business needs scalable voice/video/data networks; more applications and users mean more threats, so network security is central.

- CIA is the goal set: conceal in storage and transit; authorize changes (including against accidental corruption); keep authorized access working (a bank that cannot serve accounts fails availability).
- A threat is potential danger; an attack is a deliberate intelligent assault. Passive attacks steal or infer information; replay, modification, and DoS produce unauthorized effects; snooping is stopped by making data meaningless via encryption.
- Cryptography today is three mechanisms: shared-key ciphers, public/private-key (RSA-style) algorithms, and iterated hashes (MD2/4/5; SHA from NIST). MD5's 128-bit digest is too small.
- Boundary defense: border router ACLs (ICMP, source routing, SYN, Smurf-style spoofing via ingress/egress), then firewalls (static vs stateful vs proxy), then IDS alerts and IPS auto-blocks.
- Defense in depth also needs a clear, scoped, dated, specific, understandable policy with named authority — not only tuned rule bases.

M03: Computer Network Reference Models

Learning focus

- Define a computer network and distinguish it from a distributed system (middleware / World Wide Web example).
- Classify networks by coverage: PAN, LAN, MAN, WAN, with named technologies and topologies.
- State the seven OSI-layer design principles and the PDU names from bit through APDU.
- Contrast **chained** layers (physical–network) with **end-to-end** layers (transport–application).
- List the function of each OSI layer (including MAC as a data-link sublayer).
- Describe the four-layer TCP/IP model, its ARPANET / DoD origins and design goals, key protocols, and how it differs from OSI.

Key terms

Term	Meaning in this lecture
Computer network	Autonomous computers interconnected by a single technology, able to exchange information
Distributed system	Independent computers appearing as one coherent system (often via middleware)
Middleware	Software above the OS that implements the distributed-system illusion
PAN / Bluetooth	~10 m personal interconnection; short-range radio
IEEE 802.3 Ethernet	Bus-based broadcast LAN, decentralized control, 10 Mbps–10 Gbps
IEEE 802.5 Token Ring	Ring LAN at 4 and 16 Mbps; transmit after capturing a token
FDDI	Another ring-network example

Term	Meaning in this lecture
WiMAX	Wireless MAN technology
Head-end	Central MAN point that feeds TV and Internet to homes
Subnet	WAN communication lines + switching elements/routers, typically telco/ISP owned
OSI	Open Systems Interconnection; ISO seven-layer model
Chained layers	Physical, data link, network — hop-by-hop, present on intermediate devices
End-to-end layers	Transport through application — only on end computers
TPDU / SPDU / PPDU / APDU	PDUs at transport, session, presentation, application
MAC	Medium Access Control sublayer: shared-channel access on broadcast networks
ARPANET	DoD-sponsored 1960s research network; grandfather of the Internet
TCP vs UDP	Reliable connection-oriented vs unreliable connectionless transport

Exam lines

- A network interconnects autonomous computers; a distributed system (WWW via middleware) makes them look like one machine.
- Coverage taxonomy: PAN (~10 m, Bluetooth), LAN (Ethernet bus / Token Ring / FDDI), MAN (cable TV, WiMAX, head-end), WAN (customer hosts plus telco/ISP subnet of lines and routers).
- OSI's seven layers follow explicit design principles; PDUs run bit → frame → packet → TPDU → SPDU → PPDU → APDU; lower three layers are chained, upper four are end-to-end.
- Each OSI layer has a distinct job: bits and media; frames, ACK, flow control, MAC; routing, congestion, addressing, QoS; end-to-end TPDUs and service type; sessions (dialog, tokens, sync); syntax/semantics; user protocols (HTTP, mail, news, file transfer).
- TCP/IP is the living Internet model: four layers, ARPANET/DoD goals (internetworking, surviving as long as endpoints live, flexibility), IP plus TCP/UDP plus Telnet/FTP/SMTP/DNS/NNTP/HTTP.
- Shared idea of a protocol stack and transport-to-applications split; OSI has more layers and a cleaner service/interface/protocol split; TCP/IP puts connectionless IP below and lets the user choose TCP or UDP above.

M04: Computer Network Reference Models — TCP/IP Layers Continuation

Learning focus

- Trace TCP/IP from ARPANET / DARPA (Kahn and Cerf, NCP) to the IETF-maintained Internet protocol suite.
- Map the four TCP/IP layers (and the five-layer hybrid that adds physical) onto OSI, including chained vs end-to-end layers.

- List link-layer tasks (MAC, LLC) and named link protocols: IEEE 802 family, ATM, ARP, NDP, LLTD, PPP.
- State IP's three functions and the internet-layer protocols IP, ICMP, IPsec/IKE, IGMP, and OSPF.
- Contrast TCP and UDP (RFCs, reliability, sockets/ports) and name DCCP, RSVP, TLS/SSL, and SCTP.
- Identify application-layer protocols with their RFCs: Telnet, FTP, SMTP, HTTP, RPC, DNS, SNMP, RTP.

Key terms

Term	Meaning in this lecture
Internet protocol suite / TCP/IP	Model and protocols of the Internet; named for TCP and IP
DoD model	Alternate name because ARPANET was DoD-sponsored
IETF	Body that maintains TCP/IP and related models
DARPA	Agency whose late-1960s R&D produced the suite
NCP	ARPANET Network Control Program; Cerf was its developer
Five-layer hybrid	Four TCP/IP layers plus an explicit physical layer 1
MAC / LLC	Shared-medium access vs flow control, link management, error handling, packet sizing
ATM	Fixed-size cells; mix real-time audio/video with regular data
ARP / NDP	IP→MAC mapping for IPv4 vs IPv6 neighbor discovery
LLTD	Microsoft proprietary topology / QoS discovery
PPP	ISP dial-up over a telephone line between two nodes
IP	Unreliable datagram forwarding; fragmentation; IPv4 and IPv6
ICMP	Errors and diagnostics (echo) sitting on IP
IPsec / IKE	Authenticate/encrypt IP datagrams; key exchange
IGMP	IP multicast group administration
OSPF	Link-state IGP inside one AS; called the most widely used Internet routing protocol here
Socket	IP address + port = unique connection endpoint
RFC 793 / RFC 768	TCP / UDP standards
DCCP / RSVP / SCTP	Congestion-aware datagrams; resource reservation; scalable stream TCP-like transport
TLS / SSL	Encrypt TCP segments; keys, auth, encryption, signatures

Exam lines

- TCP/IP is a practice-first, IETF-maintained DoD/ARPANET descendant: Kahn and Cerf hid network differences behind a common internet protocol and moved reliability from the network to the **host** (replacing NCP).
- Four layers (or five if physical is drawn separately): link and internet are chained; transport and application are end-to-end process communication. OSI protocols were designed first and are unused; TCP/IP dominates.

- Link layer is local (no routers): IEEE 802 (Ethernet, Token Ring, 802.11), ATM cells, ARP (IPv4) vs NDP (IPv6), Microsoft LLTD, and ISP PPP dial-up.
- Internet layer: hierarchical addressing, hop-by-hop routing, congestion control — IP (unreliable, fragmenting, v4/v6), ICMP, IPsec+IKE, IGMP multicast, OSPF inside an AS.
- Transport: ports and sockets; TCP (RFC 793, reliable bidirectional) vs UDP (RFC 768, fast, lossy, unordered); also DCCP, RSVP, TLS/SSL, SCTP.
- Application is the OSI 5–7 interface: Telnet, FTP, SMTP, HTTP, RPC, DNS, SNMP, RTP, several explicitly tied to RFCs and to client–server over TCP.

M05: Protocol Stack

Learning focus

- Distinguish a protocol **suite** (definition) from a protocol **stack** (software implementation) and state why layered modules exist.
- Recite the five OSI layering principles (including ISO 7498) and describe encapsulation/decapsulation (headers, Layer-2 trailer, physical signals).
- List services of each of the seven OSI layers, with named mechanisms (baud rate, line codes, CSMA/CD, IP/ICMP/IGMP/IPsec, TCP vs UDP, session beans, etc.).
- Describe the four TCP/IP layers and early applications Telnet, FTP, SMTP, and DNS (including the Yahoo name-to-address example).
- State the lecture's listed merits and demerits of TCP/IP.

Key terms

Term	Meaning in this lecture
Protocol suite	Definition of the protocols
Protocol stack	Software implementation of the suite; hierarchy of layers
ISO 7498	Identifier of the OSI model standard
D7…D1	Data units at OSI layers 7 down to 1
Baud / symbol rate	Symbol (or pulse) changes per second on the medium
Modem	Device that both modulates and demodulates
Line encoding	Unipolar, polar, bipolar, Manchester patterns of voltage/current/photons
Simplex / half / full duplex	One-way; either direction but not both at once; both at once
Frame	DLL PDU: link header + packet; trailer typically only here
ARQ	Retransmit specified frames when errors are detected
IEEE 802.2 LLC	Common LAN logical-link protocol
CSMA/CD vs token passing	Ethernet MAC vs Token Ring MAC
Slow start	Congestion avoidance that starts (or restarts after loss) at low rate
UDP-Lite	Lightweight UDP variant; still connectionless, no inherent congestion control

Term	Meaning in this lecture
Session beans	Session-scoped objects deleted when the web session disconnects
Network virtual terminal	Emulated terminal so a remote host accepts a logon
www.yahoo.com → 204.71.177.71	DNS example given in the lecture

Exam lines

- A suite is the specification; a stack is the running layered software, often split into media / transport / applications with two OS interfaces (e.g. TCP/IP ↔ Ethernet, browser ↔ TCP/IP).
- OSI (ISO 7498) uses seven abstraction layers with explicit design rules; data grows headers (and a layer-2 trailer) going down, becomes an EM signal, and is unwrapped going up.
- Physical: bits, baud, connectors, radio/IR, modulation/modems, line codes, bit sync, point-to-point vs multipoint, topology, duplex modes.
- Data link: frames, hardware addresses, flow control, ARQ, LLC (802.2), MAC (CSMA/CD vs token passing).
- Network: logical IP addresses, routing, datagrams, fragmentation, ICMP/IGMP/IPsec/IPv4/IPv6
- Transport PDUs traverse the network unmodified; TCP is a reliable connection; UDP/UDP-Lite are cheap and connectionless — applications must not blast at line rate. Ports multiplex (OSI would put that in session).
- Session (auth, authorization, restoration, checkpointing; session beans, lip-sync, live TV), presentation (translate, encrypt, compress), application (mail, virtual terminal, directories, FTAM).
- TCP/IP: underspecified host-to-network, connectionless IP, TCP or UDP (full or half duplex), Telnet/FTP/SMTP/DNS on top. Merits: independence, scale, client–server, many routing protocols, can connect two computers. Demerits: transport does not always guarantee delivery, poor reuse in other applications, hard protocol replacement, fuzzy service/interface/protocol split.

M06: Transmission Control Protocol

Learning focus

- Place TCP in the Cerf–Kahn 1974 internetworking paper and in RFC 793 / 1122 / 1323.
- Explain reliable stream delivery: segments, acknowledgements, timers, retransmission, and nesting TPDU ⊂ packet ⊂ frame.
- Define sockets, TSAPs/ports (including well-known ports below 1024 and IANA), and TCP's full-duplex, point-to-point, byte-stream service.
- Walk the 20-byte TCP header field by field (ports, seq/ack, data offset, flags, window, checksum, urgent pointer, options).
- Use socket primitives and the three-way handshake (plus simultaneous-open collision and FIN release / two-army timers).

- Apply variable-size sliding-window flow control, including a zero window and the one-byte probe that prevents deadlock.

Key terms

Term	Meaning in this lecture
Transmission control program (1974)	Cerf–Kahn component later split into TCP + IP
RFC 793 / 1122 / 1323	TCP definition; clarifications and bug fixes; extensions
Segment	TCP unit of transmission; IP encapsulates it for delivery
TPDU	Transport protocol data unit nested in packets then frames
Socket / TSAP	IP + 16-bit port; port is TCP's transport service access point
Well-known ports	Numbers below 1024 reserved for standard services; IANA assigns
Byte stream	Message boundaries not preserved; writes may coalesce or split
20-byte header	Fixed TCP header; options extra; max data 65495 with 20-byte IP
ACK number	Next byte expected, not last byte received
SYN/ACK/FIN/RST/PSH/URG	Connection setup, valid ACK, close, reset, push, urgent offset
Sliding window	Variable-size; WIN says how many bytes may be sent from ACK point
Pseudo-header	Included in the TCP checksum along with header and data
Three-way handshake	SYN; SYN+ACK; ACK (seq X / Y, ACK X+1 / Y+1)
Two-army problem	Connection-release uncertainty; solved with 2× MPL timers
Zero window	Sender stops except urgent data or a 1-byte window probe

Exam lines

- TCP grew out of the 1974 Cerf–Kahn transmission control program, then split from IP; it is specified in RFC 793 with later fixes (RFC 1122) and extensions (RFC 1323).
- It is a reliable, ordered **byte stream** with ACKs, retransmission timers, and segments carried inside IP packets inside frames — not a message stream, and not multicast/broadcast.
- Endpoints are sockets (IP + port/TSAP); well-known ports below 1024 (FTP 20, Telnet 23, SMTP 25/587, DNS 53, HTTP 80, POP/IMAP, ISAKMP 500, RTSP 554, etc.) are IANA-assigned.
- The 20-byte header numbers every byte, advertises a window, checksums a pseudo-header, and uses six flags for urgent data, valid ACKs, push, reset, setup (SYN), and teardown (FIN).
- Servers socket/bind/listen/accept; clients socket/connect; close is symmetric. Setup is a three-way handshake (RST if nobody listens); simultaneous open can collide. Teardown is two FINs (often four segments, sometimes three) plus timers against the two-army problem.
- Flow control is a variable window independent of data-link-style ACK coupling: a full 4K buffer can advertise WIN=0; urgent data and a one-byte probe still get through so a lost window update cannot deadlock.

M07: Data Link Layer

Learning focus

- Place the data link layer (DLL / Layer 2) above the physical layer and split it into LLC and MAC.
- List DLL functions: framing, hardware addressing, error control, flow control, and multi-access (CSMA/CD).
- Describe four framing methods: byte count, flag bytes with byte stuffing, flag bits with bit stuffing (01111110), and physical-layer coding violations (Manchester / IEEE 802).
- Contrast stop-and-wait with sliding-window flow control, including utilization U = 1/(1+2a).
- Explain ARQ error control: stop-and-wait ARQ, Go-Back-N ARQ, and selective-repeat ARQ.

Key terms

Term	Meaning in this lecture
DLL / Layer 2	OSI layer above physical; hides hardware; frames for upper layers
Collision domain	Shared-link setting that makes DLL work harder
LLC	Point-to-point sublayer: protocols, flow control, error control
MAC	Who uses the broadcast medium; CSMA/CD for shared access
NIC	Network interface card / adapter (e.g. Ethernet) implementing DLL in hardware
Segmentation / encapsulation	Split network-layer stream; add header and frame flags
Byte stuffing	Insert ESC before accidental flag bytes in the data
Bit stuffing	After five 1s in data, insert 0; flag is 01111110
Manchester / coding violation	1→10, 0→01; illegal physical codes delimit IEEE 802 frames
Stop-and-wait	One data frame, then wait for ACK; U = 1/(1+2a)
a	Propagation time / transmission time
Sliding window	Numbered frames; window up to 2^K−1; used by TCP (kernel buffer)
ARQ	Automatic repeat request after error detection
Stop-and-wait ARQ	ACK/NAK plus timer; duplicates discarded; small buffers, poor on long delay
Go-Back-N	Continuous send; on NAK/timeout resend that frame and all later ones; max window 2^K−1
RR	Receive Ready piggyback (e.g. RR1)
Selective repeat	Retransmit only lost/NAK frames; receiver resequences from storage

Exam lines

- Layer 2 sits on the physical bit pipe, hides the hardware, and is hardest on a shared collision domain; LLC (point-to-point protocols, flow, errors) and MAC (who speaks on the broadcast medium) split the work, often in an Ethernet NIC.
- Packets become frames with address, checksum, and control fields; four ways to mark frame edges are count, byte-stuffed flags, bit-stuffed 01111110, and Manchester coding violations (IEEE 802).
- Speed mismatches (Pentium vs 80386, Ethernet vs a 28.8 kbps modem) require flow control: stop-and-wait is simple but idle on long-delay paths (U = 1/(1+2a), bad for satellite); sliding windows of size at most 2^K−1 (and TCP's kernel buffer) keep multiple frames in flight and can cumulative-ACK.
- Errors are recovered by ARQ: stop-and-wait ARQ (ACK/NAK, timer, drop duplicates) is simplest and least efficient; Go-Back-N is continuous and rewinds to the bad frame (window capped at 2^K−1 so piggybacked RR cannot be ambiguous); selective repeat is most efficient and most complex because only lost frames are resent and the receiver must reorder.

M08: Medium Access Control Protocols

Learning focus

- Place the MAC sublayer in the OSI data-link layer and list its basic functions.
- Contrast static channel allocation (FDM) with dynamic allocation and state the LAN assumptions that dynamic methods need.
- Distinguish random-access (ALOHA, CSMA, CSMA/CD) from controlled-access (polling, token passing, slotted ring) methods.
- Compute Pure ALOHA and slotted ALOHA throughput and explain the vulnerable period in each.
- Describe nonpersistent, 1-persistent, and p-persistent CSMA, and the CSMA/CD collision procedure used on early Ethernet.

Key terms

Term	Meaning in this lecture
MAC sublayer	Bottom of the data-link layer; IEEE media-access, error detection, station addressing; serves LLC, uses the physical layer
LLC	Logical Link Control; MAC's user above
Multiple access	Many stations share one broadcast medium
FDM	Static split of bandwidth into fixed frequency partitions
Collision	Overlapping transmissions that garble the signal
Carrier sensing	Electrical sense that the channel is busy or free
ALOHA	Random-access protocol from the University of Hawaii (early 1970s)
Vulnerable period	Time window in which another packet can overlap and destroy a transmission
G, S	Offered load per packet time; successful throughput per slot
CSMA	Sense-before-transmit multiple access

Term	Meaning in this lecture
1-persistent CSMA	Keep sensing a busy medium, then send with probability 1 (Ethernet CSMA/CD)
p-persistent CSMA	After idle, send with probability p (CSMA/CA, Wi-Fi)
CSMA/CD	CSMA plus collision detection, jam, and backoff (early Ethernet)
10BASE5 / 10BASE2 / 10BROAD36 / 1BASE5	CSMA/CD cabling options named by Mbps, baseband/broadband, and segment length × 100 m

Exam lines

- Shared-medium LANs need MAC so simultaneous sends do not garble frames; MAC sits under LLC and above the physical layer.
- Static FDM wastes spectrum or starves users when traffic is not small and constant; dynamic methods are what LANs and WANs actually use.
- Random access (ALOHA, CSMA, CSMA/CD) versus controlled access (polling vs token/slotted ring) is the main protocol split; register insertion is obsolete.
- Pure ALOHA needs no sync but tops out near 18% ($S = Ge^{-2G}$ at $G = 0.5$); slotted ALOHA halves the vulnerable period and reaches about 36.8% ($S = Ge^{-G}$ at $G = 1$).
- CSMA listens first; 1-persistent is Ethernet/CSMA/CD, p-persistent is Wi-Fi/CSMA/CA.
- CSMA/CD aborts into a jam and backoff, and is taught as more efficient than ALOHA and plain CSMA on a bus with bursty traffic.

M09: TCP/IP Model-IP Addressing – I

Learning focus

- Trace a payload vertically down the TCP/IP stack from application to physical layer.
- State what a protocol is and when TCP versus UDP is used at the transport layer.
- Bind well-known applications to port numbers and describe a TCP segment's source and destination ports.
- Explain IPv4 dotted-decimal addressing, classful classes A–E, default masks, and network versus host bits.
- Compute network, broadcast, and gateway addresses, and state why CIDR/subnetting is introduced to cut classful waste.

Key terms

Term	Meaning in this lecture
All People Seem To Need Data Processing	Mnemonic for the seven layers from application down
Payload	Application data handed down (e.g. `www.google.com`)
Protocol	Set of rules for a line of action (TCP or UDP at transport)
Segment	Transport PDU: payload plus source and destination ports
Port	16-bit application identifier; 65,536 possible values
Well-known ports	Below 1024; not used as random source ports
Dotted-decimal	IPv4 written as four octets 0–255
Subnet mask	Marks which IP bits are network vs host
Unicast	One machine talking to one other machine
Network address	IP AND mask; host bits zero
Broadcast address	All host bits one
Gateway	Router address used to reach another network
CIDR	Classless inter-domain routing; subnetting to reduce classful waste

Exam lines

- Layers talk only **down the stack**; the browser payload becomes a TCP segment (ports) then an IPv4 packet (addresses).
- TCP is taught as the default for HTTP and mail; UDP is the other transport choice.
- HTTP listens on **80**; sources pick a port **above 1024**; FTP 20/21, SMTP 25, SSH 22, Telnet 23.
- IPv4 is 32-bit dotted decimal; classful A/B/C (with D multicast, E reserved) waste space when a site needs only 20 hosts.
- Network = IP AND mask; broadcast = host bits all 1; a gateway (often last host address) is required to leave the network; usable hosts are $2^n - 2$.
- CIDR/subnetting uses only as many host bits as needed (five bits for 20–31 hosts); mask arithmetic and MAC addresses are the next lecture.

M10: TCP/IP Model-IP Addressing – II

Learning focus

- Finish the 20-host CIDR example: mask `255.255.255.224`, block size, network and broadcast.
- Repeat the method for 10 hosts (`/28`) and list network, broadcast, first/last host, and wasted addresses.
- Write the same prefix in subnet-mask form and in CIDR slash notation.
- Distinguish IPv4 **logical** addresses from 48-bit **MAC physical** addresses (OUI + serial).
- Use ARP (broadcast request, unicast reply) to fill a frame when two hosts are on the same network; note that off-net frames use the **gateway's** MAC.

Key terms

Term	Meaning in this lecture
CIDR	Classless inter-domain routing; slash prefix equivalent to a mask
Incomplete octet	The mask octet that is not 0 or 255; used to get block size as $256-$ that value
Block size	Number of addresses in the CIDR block (e.g. 32 for mask 224, 16 for 240)
`/28`	28 network bits = mask `255.255.255.240`
Frame	Data-link PDU with source and destination MAC
Logical vs physical	IP vs MAC (phone number vs IMEI analogy)
OEM / OUI	First three MAC bytes: manufacturer
`ipconfig /all`	Shows IP, mask, and MAC
`ping`	Reveals the current IP of a name such as `www.google.com`
ARP	Broadcast request / unicast reply to map IP → MAC on the same network
Gateway MAC	Destination MAC of an off-net frame (not the remote server's MAC)

Exam lines

- For 20 hosts, mask **`255.255.255.224`**, block **32**, network `.0`, broadcast `.31`; leftover `.21`–`.30` are waste.
- For 10 hosts, **`192.168.1.0/28`** = **`255.255.255.240`**, block **16**, broadcast `.15`, 14 usable, four leftover if only 10 are assigned.
- Network layer = host-to-host **packet**; data-link = hardware **frame** with 48-bit MACs (24-bit OEM + 24-bit serial).
- Names become IPs (`ping`); IPs become MACs (`ipconfig /all` plus ARP).
- Same-net destination MAC comes from ARP; **off-net destination MAC is the gateway**, not the server — the usual student error.
- Transport = process-to-process (ports), network = host-to-host (IP), data-link = node-to-node (MAC).

M11: TCP Connection Management and Flow Control

Learning focus

- Explain how TCP can be connection-oriented while using connectionless IP.
- Walk the three-way handshake (passive/active open, SYN, SYN+ACK, ACK) including sequence-number consumption.
- Describe SYN flooding as a denial-of-service attack and the mitigations taught (limits, filters, cookies).
- Account for bidirectional data transfer, piggybacked ACKs, PSH, and URG (including what urgent mode is not).
- Contrast three-way connection termination with four-way half-close.

Key terms

Term	Meaning in this lecture
Virtual path	TCP's connection-oriented path over connectionless IP
Passive open	Server ready to accept; does not itself initiate
Active open	Client request to connect to a particular server
SYN	Control segment that syncs ISN; consumes one sequence number; no real data
SYN+ACK	Server's reverse SYN plus ACK of client SYN; carries rwnd
ACK (handshake 3)	Usually no data; consumes no sequence number
ISN	Random initial sequence number
TCB	Transmission control block; state allocated per embryonic connection
SYN flooding	DoS via forged SYNs that exhaust server resources
Cookie	Postpone allocation until the client IP is shown valid
Piggybacking	ACK carried with data
PSH	Push: send/deliver without waiting to fill buffers
URG / urgent pointer	Mark a stream range for special treatment; not priority or out-of-order delivery
FIN	Close in one direction; consumes one sequence number if no data
Half-close	Stop sending but keep receiving (e.g. sort-then-reply)

Exam lines

- TCP is virtually connection-oriented on top of unaware, connectionless IP (retransmission and reordering are TCP's job).
- Full-duplex use needs a three-way handshake: SYN (ISN, one seq), SYN+ACK (reverse ISN + rwnd), ACK (usually no seq consumption).
- SYN flooding is DoS by fake source IPs filling TCB/timer resources; limits, filters, and cookies are the taught defenses.
- Data is bidirectional with piggybacked ACKs; PSH is for interactive immediacy (often ignored); URG only marks special bytes in order.
- Close is three-way FIN / FIN+ACK / ACK, or four-way half-close so one side can finish sending while the other still replies.

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Volume 02 — M12–M23

Information Security (CEC) · SWAYAM / CEC Information Security **notes-3** — concept exam notes (definitions & what things do). Breadth over lab depth.

M12: IPV4 & IPV6

Learning focus

- Place IPv4 as RFC 791's connectionless 32-bit packet protocol and parse every IPv4 header field taught.
- Contrast classful IPv4 ranges and default masks with CIDR (RFC 1519) using the Cambridge / Oxford / Edinburgh example.
- Explain NAT and the RFC 1918 private ranges.
- List IPv6's 128-bit notation, header fields, address compression, and six extension headers.
- State which IPv4 header fields disappear in IPv6 and why.

Key terms

Term	Meaning in this lecture
RFC 791 / RFC 760	IPv4 spec (1981) replacing the 1980 definition
IHL	Header length in 32-bit words; 5–15
DSCP / ECN	Diffserv class; congestion signal without drops
DF / MF	Don't fragment; more fragments
Classful addressing	Historic A–E split; no longer used
CIDR (RFC 1519)	Variable blocks, class-free, aligned on block size
NAT	One public IP; rewrite internals at the border
RFC 1918	Private ranges 10/8, 172.16/12, 192.168/16
Flow label	IPv6 pseudo-connection / QoS-related marking

Term	Meaning in this lecture
Hop limit	IPv6 TTL-as-actually-used
`::`	Compression of one or more zero 16-bit groups

Exam lines

- IPv4 is connectionless, 32-bit, RFC 791; the header is 20 bytes plus up to 40 bytes of options, with DSCP/ECN, fragmentation (DF/MF/offset), TTL, protocol, and header-only checksum.
- Classful A–E is historical; CIDR hands out aligned variable blocks (Cambridge 2048, Oxford 4096 on a 4096 boundary, Edinburgh 1024).
- NAT plus RFC 1918 private space, often with a firewall, keeps internal 10.x addresses off the public Internet.
- IPv6 is 128-bit, 40-byte fixed header, seven fields, better options/security/QoS; addresses compress with dropped leading zeros and `::`.
- IPv6 drops IHL, protocol, fragmentation fields, and checksum; TTL is renamed hop limit; six extension headers (including AH and ESP) carry what IPv4 stuffed into options and flags.

M13: UDP — User Datagram Protocol

Learning focus

- Place UDP between applications and IP as a connectionless, unreliable, low-overhead process-to-process protocol.
- Parse the 8-byte user-datagram header and the hexadecimal dump example (ports, length, data).
- List UDP services: well-known ports, no connection, no flow/window, checksum with pseudo-header, no congestion control.
- Trace encapsulation/decapsulation, queuing, and multiplexing/demultiplexing.
- Describe the five-component UDP “package” (control-block table, input queues, control-block / input / output modules).

Key terms

Term	Meaning in this lecture
User datagram	UDP packet; 8-byte header plus data
Ephemeral port	Client port chosen by UDP/OS (vs well-known server ports)
Socket	IP address + port
Pseudo-header	IP-derived fields (including protocol 17) included in the UDP checksum
Silent drop	Checksum failure with no NAK to the sender
Protocol 17	IP protocol number for UDP
Multiplexing	Many processes → one UDP at the sender
Demultiplexing	One UDP → many processes at the receiver

Term	Meaning in this lecture
Control-block table	Open-port state, PID, port, queue number
ICMP on unknown port	Input module's response when no table entry matches

Exam lines

- UDP is a thin, connectionless, unreliable transport: ports for process-to-process delivery, almost no control, silent drop on error.
- Header is always 8 bytes (src, dest, length, checksum); example `CB84/000D/001C/001C` is client port 52100 to daytime (13) with 20 data bytes.
- No connections, numbering, flow window, or congestion control; checksum (with pseudo-header) is the only error tool and is optional versus a bare datagram protocol.
- Encapsulation sets IP protocol 17; decapsulation peels physical → link → IP → UDP and passes the sender socket up for a reply.
- Features that look like weaknesses (no setup, no retry) are taught as advantages for short exchanges and for not feeding congestion.
- The sketched package: control-block table, per-process input queues, modules for bind, receive (ICMP if no port), and send.

M14: IPSec Security Protocol

Learning focus

- Define IPsec as a network-layer framework (not a single protocol) for authenticating and encrypting IP packets.
- Name architecture pieces from RFC 2401: security associations, AH, ESP, transport vs tunnel mode, and IKE.
- Contrast transport mode (payload protected; IP header intact) with tunnel mode (whole packet wrapped; used for VPNs).
- Parse AH and ESP headers/trailers, ICV, SPI, and sequence numbers for anti-replay.
- State IKE's role (RFC 2409): mutual authentication, SA setup, aggressive/main modes, and three authentication methods.

Key terms

Term	Meaning in this lecture
IPsec	Network-layer suite: authenticate and encrypt each IP packet; useful for VPNs and dial-up without host software changes
RFC 2401	IPsec architecture (SA, AH, ESP, modes, IKE)
SA	Security association: algorithms and parameters for AH/ESP
Transport mode	Crypto endpoints = communication endpoints; payload protected, IP header intact

Term	Meaning in this lecture
Tunnel mode	Whole packet wrapped in a new IP header; VPNs; at least one crypto endpoint is a gateway
AH	Integrity and origin authentication; optional anti-replay; no privacy
ESP	Encryption (privacy) plus optional authentication
SPI	32-bit index which, with dest IP, selects the SA
ICV	Integrity check value from a keyed hash
IKE (RFC 2409)	Mutual auth and SA/key setup; main vs aggressive; PSK / public-key encrypt / public-key signature

Exam lines

- IPsec secures IP itself (not the application), for host–host, gateway–gateway, or gateway–host, including VPNs, without reconfiguring every PC.
- Architecture is SAs plus AH and/or ESP in transport or tunnel mode, with IKE for keys.
- Transport keeps routing headers visible; tunnel hides the original packet behind gateway addresses and builds VPNs.
- AH authenticates (hash + SA key + ICV, SPI, anti-replay sequence); it does not give confidentiality — that is ESP.
- ESP splits header (SPI, seq), encrypted payload, trailer (pad, pad length, next header), and optional ICV after encryption for block ciphers and post-encrypt authentication.
- IKE (RFC 2409) negotiates SAs with main or aggressive mode and three auth methods; the lecture treats full crypto detail as out of scope.

M15: Network Topologies

Learning focus

- Define a network (nodes, links, distributed processing) and the five criteria: performance, consistency, reliability, recovery, security.
- Distinguish point-to-point from multipoint (spatially vs time-shared) connections.
- Separate physical topology from logical topology.
- Identify mesh (full and partial), star (passive/active/intelligent hub), bus, ring (token), and tree (expanded star), with the advantages and disadvantages taught for each.
- Use those trade-offs to match a topology to a network plan.

Key terms

Term	Meaning in this lecture
Node	Computer, printer, or other send/receive device
Distributed processing	Split a task across multiple machines

Term	Meaning in this lecture
Transit vs response time	Message travel time vs inquiry-to-answer delay
Throughput	Successful messages per unit time (file size / time)
Four delay components	Processing, queuing, transmission, propagation
Point-to-point / multipoint	Dedicated pair vs shared link (spatial or time-shared)
Physical vs logical topology	Cable/device layout vs how data actually flows
Full mesh	$n(n-1)/2$ duplex links; backbone use
Passive / active / intelligent hub	Splitter vs regenerator vs path-selecting (router/bridge/switch)
Terminator / barrel connector	Stop bus reflections; extend the bus
Token	Permission to send on a ring
Tree / expanded star	Stars joined by a bus backbone

Exam lines

- Networks are linked nodes doing distributed work; they are judged on performance (throughput vs delay), consistent response, reliability, recovery from backups, and security (antivirus, firewalls, backups, unplug when idle).
- Links are point-to-point (dedicated, including IR remote or satellite) or multipoint (spatial or turn-taking share).
- Physical layout can differ from logical data flow; the taught shapes are mesh, star, bus, ring, and tree.
- Full mesh is $n(n-1)/2$ duplex links and is for backbones; partial mesh cuts cost; wireless often meshes with flooding or routing.
- Star is easy to grow and troubleshoot but dies with the hub; bus is cheap for small LANs but broadcasts to everyone and dies with the backbone; ring is ordered and collision-light via tokens but one break stops the ring; tree scales star+bus until the stem cable fails.

M16: Animated Cursor Vulnerability — Proof of Concept I

Learning focus

- Restate **CVE** (Common Vulnerabilities and Exposures) and why a **2007 Windows animated-cursor** flaw is still taught.
- Distinguish a **client-side attack** from traffic a **perimeter firewall** can filter.
- Define **remote code execution (RCE)** as unauthorized code running on someone else's machine.
- Explain at a concept level that Windows loads variable-size **`.ANI`** cursors via **`user32.dll`**, and that a malformed cursor can cause a **buffer overflow**.

- Name the lecture's defensive close: **patch** legacy systems; study **client-side attacks** and **buffer overflow**.

Key terms

Term	Meaning in this lecture
CVE	Common Vulnerabilities and Exposures — catalog of named flaws
Animated cursor / `.ANI`	Variable-size Windows pointer animation, loaded through `user32.dll`
RIFF	File header that identifies an animated cursor
Network firewall	Perimeter filter for inbound traffic; does not stop a user who goes out to the web
Client-side attack	Compromise that starts because the insider visits a hostile page
Remote code execution	Running code on another machine without authorization
EIP	Instruction pointer; overflow target in the demonstration
Patch	The stated way to avoid the flaw on systems that still run the old loader

Exam lines

- Perimeter firewalls do not stop **client-side** browsing of a page that hosts a malicious cursor.
- The 2007 animated-cursor bug is a **buffer overflow in the Windows cursor loader** (`user32.dll`) that can yield **RCE** on **unpatched XP / 2000 / Vista**.
- A page need only **assign a CSS cursor URL** to a crafted **RIFF `.ANI`**.
- Overflow that **overwrites EIP** is the diagnostic that attacker-controlled data became the next-instruction address.
- **Patch** legacy systems; study client-side attacks and buffer overflow.

M17: Animated Cursor Vulnerability — Proof of Concept II

Learning focus

- Continue the animated-cursor demonstration conceptually: after EIP is attacker-controlled, execution can be redirected using instructions **already mapped** in system DLLs.
- Contrast **bind** vs **reverse** remote-shell models as firewall-relevant ideas (inbound listen vs victim-initiated outbound).
- Recite the **defensive lessons actually taught**: browse with **least privilege**, treat unexpected email links as client-side lures, **patch / hotfix** from **cve.mitre.org**, and see **malware / ransomware** as the next threat category.

Key terms

Term	Meaning in this lecture
Mapped DLL	Code already in the process (`user32` and others) that overflow control can be aimed at
Bind vs reverse shell	Inbound listen vs victim-initiated outbound — the latter matches allowed web traffic
Least privilege	Do not browse as Administrator; the attacker inherits the logged-in token
cve.mitre.org	Public CVE catalog; patch/hotfix against listed issues
Malware / ransomware / CryptoWall	Harmful software; CryptoWall encrypts files and extorts payment
Malvertising	Advertisement channel used to push malware

Exam lines

- After EIP is untrusted, execution can be steered using **code already in system DLLs**; the defender's answer is still **patch the loader**.
- **Reverse** connections piggyback on **user-initiated outbound** paths that firewalls already allow for the web.
- **Administrator browsing** makes a client-side bug a full compromise; **least privilege** is the taught countermeasure on the endpoint.
- **Lottery-style email links** are the same client-side pattern: the user **asks** to fetch the page.
- Track flaws on **cve.mitre.org** and **patch / hotfix**; unpatched loaders on XP/2000/Vista (CVE-2007 animated cursor) were the worked example.
- The course next frames **malware** (including **ransomware / CryptoWall**) as a fourth companion to software, hardware, and firmware.

M18: Electronic Mail

Learning focus

- Define **email** as store-and-forward digital messaging and list its **five basic functions**.
- Split an email system into **user agents (UA)** and **message transfer agents (MTA)** and match them to **RFC 822 / MIME** versus **SMTP**.
- Describe **RFC 5322** headers (To, Cc, Bcc, From, Sender, Received, Return-Path, and UA fields) and **MIME** headers and body hierarchy.
- Trace **SMTP** submission, **MX** lookup, and the **MAIL / RCPT / DATA** transaction.
- Contrast **POP3**, **IMAP**, and **MAPI** as message-reading protocols.

Key terms

Term	Meaning in this lecture
Store-and-forward	Servers accept, forward, deliver, and store; parties need not be online together
User agent (MUA/UA)	Local compose/read program
MTA / MSA / MDA	Transfer, submission, and delivery agents in the SMTP path
MIME	RFC 2045–2049: types, encodings, multipart body tree
RFC 5322	Current Internet message format (via RFC 822 → 2822 → 5322)
SMTP / ESMTP	RFC 821 then RFC 5321 (2008); MAIL, RCPT, DATA on TCP
MX record	DNS mail exchanger for the domain after `@`
LMTP	Local Mail Transfer Protocol — SMTP derivative for MDA hop
POP3	Port 110; download-and-delete; AUTH / Transaction / Update
IMAP4	Port 143; server-resident mail; multiple mailboxes
MAPI	Microsoft Outlook–Exchange proprietary protocol

Exam lines

- Modern email is **store-and-forward** with five functions: compose, transfer, report, display, dispose.
- **UAs** format with **RFC 822/5322 + MIME**; **MTAs** move mail with **SMTP**.
- MIME added **typed, hierarchical bodies** so mail is more than ASCII English.
- SMTP submission is **587** (still often **25**); routing uses **DNS MX**; a transaction is **MAIL, RCPT, DATA**.
- **POP3** pulls and typically **deletes** on the server; **IMAP** keeps the **server copy as truth** and supports folders and partial fetch; **MAPI** is Outlook/Exchange only.

M19: Email Security Protocols

Learning focus

- Explain why ordinary email is compared to a **postcard**, not a sealed letter, including **transit and at-rest** inspection and **cross-border** legal complexity.
- Describe **PGP** (and **OpenPGP**) as hybrid encryption plus signatures, compression, and the taught **RSA key-length** choices.
- Outline **PEM** (RFC 1421–1424): canonicalization, hash, DES, one-time keys, **X.509** hierarchy — and **why it was never deployed**.
- State **S/MIME** services and its **multiple trust anchors** versus PEM's single root.
- Compare **S/MIME v3** and **OpenPGP** as the two systems the lecture treats as currently used.

Key terms

Term	Meaning in this lecture
Postcard model	Email visible to every relay; unlike a sealed envelope

Term	Meaning in this lecture
PGP	Zimmermann 1991; IDEA + RSA + MD5 + ZIP + signatures; free, multi-OS
Session key	One-time symmetric key; RSA wraps it for the recipient
Alien-strength RSA	2048-bit; recommended for all PGP use here
OpenPGP	IETF RFC 4880 (from RFC 1991 / 2440); Camellia RFC 5581; ECC RFC 6637
PEM	RFC 1421–1424; DES; X.509 single-root CA; CRLs; unused
Web of trust	PGP's PKI answer to PEM's central root
IPRA / PCA	Policy authorities in the PEM certificate story
MOSS	MIME Object Security Services — post-PEM MIME security effort
S/MIME	RFC 2632–2643; MIME + crypto; multiple trust anchors
Non-repudiation	S/MIME service listed with authentication, integrity, secrecy

Exam lines

- Unprotected email is a **postcard** on untrusted servers; cryptography is necessary but **jurisdiction** makes privacy messy.
- **PGP** hybridizes **IDEA** (bulk) and **RSA** (tiny hash and session-key blobs), with **ZIP**, **MD5** signatures, and a strong push to **2048-bit** RSA.
- **OpenPGP** (RFC 4880) is the interoperable IETF form; both parties must run it; it also protects **stored** files.
- **PEM** matched PGP's goals with a **single X.509 root** and **DES**, and **died** on root politics, patent fees, and MIME.
- **S/MIME** keeps PEM-like services but allows **many trust anchors** and lives **inside MIME**.
- **S/MIME v3** and **OpenPGP** both secure mail; **certificates and encodings differ**, so they do not interoperate.

M20: WWW Security

Learning focus

- Explain why the **WWW** as a **client–server TCP/IP** application creates security problems that traditional publishing did not.
- Group Web threats as **passive vs active** and as **integrity, confidentiality, authentication, and DoS**, with the lecture's consequences and countermeasures.
- Place Web security in the stack: **IPsec** (network), **SSL/TLS** (transport), **SET / PGP / S/MIME** (application), plus **firewalls**.
- Distinguish IPsec **transport vs tunnel** mode and **AH vs ESP**.
- Outline **SSL/TLS**'s four protocols and **SET**'s card-payment properties.
- Classify **packet-filter** (static/stateful) and **proxy / application-gateway** firewalls, including the lecture's filter-table examples.

Key terms

Term	Meaning in this lecture
Passive vs active Web attack	Eavesdrop / read restricted data vs impersonate, alter transit, deface
IPsec transport / tunnel	Protect transport payload vs wrap the entire original IP packet
AH vs ESP	Integrity + source auth vs those plus confidentiality
SSL / TLS	Security just above TCP; Record, Handshake, Change Cipher Spec, Alert
HTTPS	HTTP carried in SSL/TLS
SET	Mastercard/Visa card-payment security; DES + RSA + SHA-1; merchant never sees PAN
Packet filter	IP/port/protocol table; static router or stateful appliance
Application gateway	Proxy that inspects URLs/content, e.g. HTTP

Exam lines

- The Web is a **two-way, highly visible, complex** client–server system; a hacked server can **pivot** into the rest of the site.
- Threats are **integrity, confidentiality, authentication, and DoS**; DoS is called out as **hard to stop**.
- Put crypto at **IP (IPsec)**, **above TCP (SSL/TLS / HTTPS)**, or **in the app (SET, PGP, S/MIME)**.
- SET is **not a payment network**; it **hides the card number from the merchant** and **pins algorithms**.
- Firewalls **filter or proxy**; application policy that depends on **who** is on port 80 needs a **proxy**, not just a packet table.
- Technical controls still require **user awareness**.

M21: Mobile Code Security

Learning focus

- Define the **mobile-code** paradigm (applets through agents), its flexibility/bandwidth benefits, and its **two threat directions** (code vs host, host vs code).
- List host-protection requirements: **authenticate origin, verify integrity, access control, semantic checks**.
- Explain host defenses taught: **sandboxing, digital shrink-wrap / Authenticode, proof-carrying code (PCC), code signing, access control / JAAS, program checking**.
- Outline **data and code protection** for agents: signatures, **hash chaining / forward integrity**, **sliding encryption**, replication, **holographic proofs + PIR**, tamper-proof hardware, **function hiding**.
- Repeat the close: host protection is **achievable**; protecting code from a **malicious host** is still **research**; **non-cryptographic** methods are generally **not enough** for the code.

Key terms

Term	Meaning in this lecture
Mobile code	Code that runs on hosts other than its origin (applets to agents)
Sandbox	Restricted environment; JDK 1.0 applets; often too limited to be useful
Authenticode / digital shrink-wrap	Producer signs; consumer verifies origin before run
PCC / VCGen	Proof-carrying code; consumer generates safety predicate, checks producer's proof
Code signing (JDK 1.1 / ActiveX)	Valid signature → full trust; else sandbox; coarse access control
JAAS	Ties **user identity** into Java access control (JDK 1.2+ model)
Forward integrity / PRAC	Hash-chain / small authenticator over agent-collected offers
Sliding encryption	Compact encryption for small data on many hops
Holographic proof + PIR	Probabilistic, tiny checks of a computation proof without revealing query bits
Function hiding	Run an encrypted function on a malicious host; decrypt the result

Exam lines

- Mobile code buys **flexibility and bandwidth** and opens **two-way** risk: **hostile applets/ActiveX** and **hostile hosts**.
- Hosts must **authenticate origin, check integrity, constrain rights, and check semantics**.
- Practical host tools: **sandbox, signing/Authenticode, PCC, JAAS-style policies**; signing alone is a **poor user decision** and an **all-or-nothing** ACL.
- Agents need **crypto** (hash chains, sliding encryption, holographic proofs, function hiding, smart cards); **replicas/voting** help integrity of results.
- Host protection is **in production reach**; **code vs malicious host** is **still research**; **crypto is required**.

M22: Transmission Media

Learning focus

- Define **transmission media** as the physical-layer path for **electrical or electromagnetic** signals and list design factors: **bandwidth, impairment, interference, number of receivers**.
- Compare **guided** media: **UTP, STP, coaxial (baseband/broadband), fiber** (step/graded multimode, single-mode), with connectors, applications, and trade-offs as taught.

- Describe **unguided** propagation (**ground, sky, line-of-sight**), **antennas**, and **satellite uplink/downlink**.
- Distinguish **radio, microwave, and infrared** bands, properties, and applications (including **IrDA**).

Key terms

Term	Meaning in this lecture
Communication channel	Synonym for transmission media at the physical layer
UTP / STP	Unshielded vs foil/braid-shielded twisted pair; RJ-45 vs STP data connector
Subscriber loop	Twisted pair from home to telephone end office
Baseband / broadband coax	One high-speed digital signal vs many analog FDM signals on CATV plant
BNC / T / terminator	Coax connectors; terminator stops reflection
Critical angle	Governs refraction vs total internal reflection in fiber
Step-index / graded-index / single-mode	Fiber propagation families
Uplink / downlink	Ground→satellite / satellite→ground
Omnidirectional vs unidirectional	Radio vs microwave antenna behavior
IrDA	Infrared device-to-device standards (75 kbps then 4 Mbps)

Exam lines

- Rate and reach are set by **bandwidth, impairment, interference, and how many receivers share the guided medium**.
- **Twisting** cancels crosstalk; **UTP** is cheap and, here, the **fastest copper**; **STP** shields better but is **costly, must be grounded, rare in Ethernet**.
- **Coax** outperforms pair on bandwidth/noise but **attenuates fast**, can **fail the whole net**, and lost the telephone backbone to **fiber**.
- Fiber uses **reflection**, **>2 Gbps**, EMI immunity; modes are **multimode step, multimode graded, single-mode**.
- Wireless is **radio (omni, multicast), microwave (LOS unicast), infrared (in-room, IrDA)**; antennas and **licensed bands** matter.

M23: Ethernet and Fast Ethernet

Learning focus

- Place **Ethernet** historically (Xerox, commercial 1980, **IEEE 802.3** in 1983) and name the four **10 Mbps** cabling families.

- Map **LLC vs MAC**, list **IEEE 802.3 frame fields**, and walk **CSMA/CD** including **jam**, **binary exponential backoff**, and **why padding to 64 bytes** exists.
- Explain **Fast Ethernet (802.3u)** as the same rules with **bit time 100 ns → 10 ns**, and why **cable length** shrank instead of **minimum frame size**.
- List Fast Ethernet **PHY** types (T4, TX, FX, SX, BX, LX10) and the **MII / PCS / PMA / PMD** stack.

Key terms

Term	Meaning in this lecture
IEEE 802.3 / 802.3u	Ethernet (1983); Fast Ethernet (June 1995)
10BASE5 / 2 / T / F	Thick coax 500 m; thin 185 m; TP hub 100 m; fiber 2 km
Vampire tap	10BASE5 pin into the coax core
LLC / MAC	Network-protocol handoff vs shared-media access and MAC addresses
CSMA/CD	Sense, send, detect collision, jam, backoff
Binary exponential backoff	Wait 0…2^i−1 slots; cap 1023 after 10; fail after 16
Slot time / 2T	Collision window; min frame **64 bytes** so sender still transmitting when jam returns
100VG-AnyLAN	802.12 demand-priority alternative that did not succeed
MII / PCS / PMA / PMD	Fast Ethernet MAC–PHY stack; autonegotiation in PCS

Exam lines

- Ethernet grew from Xerox **3 Mbps** coax to **IEEE 802.3** and displaced Token Ring, FDDI, and ARCNET on wired LANs.
- Shared coax (**10BASE5/2**) dies on **one cable fault**; **10BASE-T** hubs and **10BASE-F** fiber (noise, distance, **harder tapping**) were the answers at 10 Mbps.
- MAC is **CSMA/CD** with **jam**, **binary exponential backoff**, and a **64-byte floor** so collisions are **heard in 2T**.
- Fast Ethernet is the **same frames and MAC** with **10 ns** bits and **~100 m** copper reach, standardized as **802.3u**; **100VG-AnyLAN** lost.
- Copper PHYs are **T4 (Cat 3)** and **TX (Cat 5, full duplex)**; fiber PHYs cover **short, long, bidirectional-single-fiber, and 10 km** plants, glued to MAC by **MII + PCS/PMA/PMD**.

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Volume 03 — M24–M35

Information Security (CEC) · SWAYAM / CEC Information Security **notes-3** — concept exam notes (definitions & what things do). Breadth over lab depth.

M24: Ethernet Security

Learning focus

- Trace Ethernet from ALOHAnet and Xerox CSMA/CD through the DIX 1.0 spec to IEEE 802.3, and name the frame as the on-wire unit.
- State why Ethernet was designed without security, and list the five attack classes the lecture uses (access, confidentiality, integrity, DoS, system security).
- Contrast router-based segmentation, access control (including IEEE 802.1Q / 802.1X), higher-layer crypto, and monitoring.
- Describe IEEE 802.1AE MACsec: SecTAG and ICV placement, SA/SAK/PN/SCI, Protect/Validate, and replay protection.

Key terms

Term	Meaning in this lecture
Frame	Ethernet's on-wire data unit; only one node should send one at a time
CSMA/CD	Carrier sense, multiple access, collision detection — Ethernet's classic shared-medium rule
IEEE 802.3	LAN/MAN standard grown from DIX Ethernet 1.0 (1980) and standardized in 1983
VLAN hopping	Injecting VLAN-tagged frames (or abusing VLAN management) to reach VLANs the port should not join

Term	Meaning in this lecture
STP TCN flood	DoS by flooding Spanning Tree topology-change notifications or other STP control
IEEE 802.1Q / QinQ / PVLAN	VLAN tagging, double tagging, and private-VLAN isolation (hosts only talk via a promiscuous router port)
IEEE 802.1X / EAP	Port authentication of a host to a switch that checks an authentication server
DHCP snooping	Bind MAC–IP–port from DHCP so ARP spoofing can be blocked, limited to what that switch can see
MACsec (IEEE 802.1AE)	Hop-by-hop encryption/integrity of Ethernet MSDUs using SecTAG, SAK, PN, SCI, and ICV
SecTAG / ICV / SAK / PN	MACsec tag, integrity check, association key, and packet number used for Protect, Validate, and replay checks

Exam lines

- Ethernet won wired LANs by simplicity, not by security; design never treated security as a first-class goal.
- Almost every attack starts with **segment access** (physical jack, rogue AP/switch, VLAN tricks, stolen switch, or a remotely owned host).
- Confidentiality, integrity (ARP/DHCP poisoning, MITM, session hijack), and STP/resource DoS are the concrete Ethernet threats taught here.
- Classic mitigation was "treat the LAN as dirty" (building + firewall) plus **IPsec/TLS**; that is costly and incomplete.
- Routers cut broadcast-domain attacks; 802.1Q/PVLAN/ACLs/802.1X limit who is on the wire; DHCP snooping hardens ARP; firewalls/IDS are secondary.
- **IEEE 802.1AE MACsec** is the Ethernet-layer protocol the lecture spends its last section on: encrypt and integrity-protect frames between authorized devices, with replay protection via PN.

M25: Gigabit Ethernet

Learning focus

- Explain why Gigabit Ethernet (1 Gbps / 1000 Mbps) was specified as a 10× Fast Ethernet successor that keeps the IEEE 802.3 frame and MAC modes.
- Map the Gigabit stack: GMII, PCS (8B/10B), PMA, PMD/MDI, and auto-negotiation of 10/100/1000 Mbps and half/full duplex.
- List the IEEE 802.3z / 802.3ab / 802.3ah (and related) physical types with media, wavelengths, and distances the lecture gives.
- Recite the basic Gigabit MAC frame fields, CSMA/CD vs full duplex, packet bursting, and **carrier extension**.

Key terms

Term	Meaning in this lecture
Gigabit Ethernet	1000 Mbps / 1 Gbps IEEE 802.3 family, 10× Fast Ethernet
1000BASE-X	IEEE 802.3z fiber/short-copper group (SX, LX, CX; ZX as vendor term)
1000BASE-T	IEEE 802.3ab Gigabit on Cat 5/5e/6 UTP
GMII	Gigabit Media Independent Interface between MAC and PHY
8B/10B	PCS coding: 8 data bits as 10-bit code groups (Fibre Channel style), 125 MHz toward PMA
Packet bursting	Half-duplex CSMA/CD enhancement: burst small packets to fill the pipe
IEEE 802.3x	Full-duplex operation and flow control used by Gigabit when not sharing the wire
Carrier extension	Extra symbols in the collision window so min/max 802.3 sizes still work at 1 Gbps distances; FCS ignores the extension
Length/Type	≤1500 → LLC length; >1500 → optional Ethernet type

Exam lines

- Gigabit exists because Fast Ethernet desktops overflowed 100 Mbps backbones; it keeps **802.3 frames** so shops need not move backbones to **ATM** or **FDDI**.
- Architecture is **MAC → GMII → PCS (8B/10B, auto-neg) → PMA (125 MHz serialize) → PMD**.
- Fiber PHYs (SX/LX/LX10/BX10/EX/ZX) and copper (CX then T) are chosen by **reach and cabling already in the wall**.
- **Full duplex** is the shipping default (up to **2 Gbps** both ways, no CSMA/CD); half duplex still exists with **CSMA/CD + burst + carrier extension**.
- Frame layout is classic Ethernet (preamble through 32-bit CRC); **carrier extension is the Gigabit-specific MAC trick** taught at the end.

M26: 10 Gigabit Ethernet

Learning focus

- State that 10 Gigabit Ethernet (10 GbE / 10G / 10 GigE) sends Ethernet frames at **10 Gbps**, first in **IEEE 802.3ae-2002**, and that it is **full duplex only** (no hubs, no CSMA/CD).
- Walk the 10G stack: RS, PCS, PMA, PMD, XGMII, MDI, and the **WAN Interface Sublayer (WIS)** for SONET-friendly WAN PHYs.
- Contrast **serial 10 Gbps** vs **parallel n-lane** (including WDM) implementations.
- Recite the unchanged Ethernet MAC frame, why the minimum is **64 octets** without carrier extension, and the LAN-fiber / WAN-fiber / copper / backplane / EPON PHYs with rates, wavelengths, and reaches.

Key terms

Term	Meaning in this lecture
10 GbE / 10G / 10 GigE	Ethernet frames at 10 Gbps; full duplex only
IEEE 802.3ae-2002	First 10G Ethernet standard (LAN/WAN fiber)
10GBASE-R / W / X	LAN fiber, WAN fiber, and copper coding families
XGMII	10 Gigabit Media Independent Interface (MAC–PHY)
WIS	WAN Interface Sublayer: simplified SONET framing between serial PCS and PMA
64B/66B	Serial LAN PCS coding; LAN PHY line rate about 10.3 / 10.3125 Gbps
HSSG	Higher Speed Study Group (March 1999) that started 10G work
10GBASE-T	IEEE 802.3an twisted-pair 10G (Cat 6 55 m; 6A/7 100 m)
10GBASE-PR	IEEE 802.3av 10G EPON PHY with PR10/PR20/PR30 power budgets

Exam lines

- 10 GbE is still Ethernet frames, but **only full-duplex switched point-to-point** — CSMA/CD and hubs are gone because half duplex at 10 Gbps is not a market.
- Stack is classic 802.3 PHY split (**PCS/PMA/PMD**) plus **XGMII** and, for WAN, **WIS/SONET**.
- Implementers choose **one 10G serial pipe** (simple) or **n slower lanes / WDM** (cheaper silicon, more jitter and optics).
- MAC frame is the familiar 64-octet-minimum Ethernet frame; **no carrier extension**.
- PHY letter soup is organized as **LAN-R, WAN-W, copper/backplane, and EPON-PR**, with **10.3125 Gbps** as the repeated serial line rate on optical LAN PHYs.

M27: ISDN

Learning focus

- Define ISDN as ITU/CCITT digital circuit- and packet-switched service over ordinary telephone-grade copper (and other media).
- Contrast BRI (2B+D = 144 kbps) with PRI (23B+D / 30B+D) and name B vs D channel rates.
- List bearer, teleservice, and supplementary services, plus TE1/TE2/TA/NT1/NT2 and R–V reference points.
- State ISDN vs ADSL differences, advantages/disadvantages, and the lecture's security conclusion (logical attacks and HTTPS matter more than the physical medium).

Key terms

Term	Meaning in this lecture
ISDN	ITU/CCITT integrated digital voice/data network on phone copper (and other media)
B channel	Bearer channel, typically 64 kbps (sometimes 56), carries user voice/data
D channel	Delta/data channel for signaling and control (16 kbps BRI, 64 kbps PRI)
BRI / 2B+D	Basic Rate Interface, 144 kbps, individual users
PRI / 23B+D / 30B+D	Primary Rate, T1-like 1.536 Mbps or E1-like 1.984 Mbps
B-ISDN	Broadband ISDN on fiber, 155–622 Mbps and beyond
Bearer / teleservice / supplementary	OSI 1–3 network functions; OSI 4–7 user services; add-ons such as Centrex and CLIP
TE1, TE2, TA, NT1, NT2	Native ISDN terminal, legacy terminal, adapter, customer NT, optional PBX
Out-of-band signaling	Setup packets on D, not ring voltage on the user channel
HTTPS	Example of the logical protection the lecture says to use on any medium

Exam lines

- ISDN digitized the local loop: **B channels for payload, D for signaling**, BRI for homes, PRI for sites that need T1/E1-scale bundles.
- Digital operation **could** have made security easier, but ISDN was **rolled out with little security design**.
- Operational wins: shared pair, ~2 s setup, out-of-band signaling, device sharing; costs and special telco service are the downsides.
- Versus ADSL: ISDN can carry voice; ADSL is data-only and often **survives local power loss**.
- Security punchline: **don't fetishize the medium**. Pick a **reputable ISP/modem**, assume **logical IP attacks**, and protect sensitive use with **HTTPS** (and similar).

M28: Stream Control Transmission Protocol (SCTP)

Learning focus

- Place SCTP as a **reliable, message-oriented** transport between UDP and TCP, aimed at telephony signaling over IP.
- Explain **multistreaming** (association vs TCP's one stream) and **multihoming** (multiple IP addresses, failover not load-share in current implementations).
- Number data with **TSN**, **SI**, and **SSN**; contrast SCTP **packets/chunks** with TCP **segments**.
- Sketch the **12-byte general header**, chunk layout, and the **four-way handshake** (INIT, INIT ACK, COOKIE ECHO, COOKIE ACK).

Key terms

Term	Meaning in this lecture
SCTP	Reliable, message-oriented transport combining UDP's boundaries with TCP's reliability, plus streams and multihoming
Association	SCTP's name for a connection (multihoming-aware)
TSN	32-bit transmission sequence number on **data chunks**
SI / SSN	Stream identifier and per-stream sequence number on data chunks
Chunk	Control or data block; several per packet; control before data
Verification tag	Per-direction association identifier (needed because of multihoming)
CRC-32	32-bit packet checksum in the 12-byte general header
INIT / INIT ACK / COOKIE ECHO / COOKIE ACK	Four-way handshake chunks
rwnd	Advertised receive window (in INIT/INIT ACK so data can ride packets 3–4)

Exam lines

- SCTP exists because **SS7-style signaling and IP telephony** need **message boundaries and reliability together**, which neither UDP nor TCP fully gives.
- An **association** can carry **many streams** (head-of-line blocking on one stream does not freeze the others) and **many IPs** (failover; **not** load-sharing in "current" implementations).
- Numbering is **chunk-based**: **TSN** globally, **SI+SSN** per stream; **control never consumes TSN**.
- The on-wire unit is a **packet**: 12-byte header (ports, verification tag, CRC-32) plus padded chunks.
- Setup is **four-way** with a **cookie**, and **user data may already appear** in the third and fourth packets.

M29: ATM Network Security Protocol

Learning focus

- Define ATM as **asynchronous time-division multiplexing** of **fixed 53-byte cells** (5-byte header + 48-byte payload) on a **SONET/ISDN-style** backbone, with a VC set up **before** data.
- Decode **UNI vs NNI** headers (GFC, VPI, VCI, PT/EFCI/AUU, CLP, HEC) and name ABR/CBR/UBR/VBR.
- List ATM Forum–style security goals (confidentiality, AAL integrity, signaling signatures, availability functions) and ATM-specific threats (**VC stealing**, traffic analysis).
- Recite QoS/traffic parameters (PCR, SCR, CLR, CTD, CDV, BT/MBS, MCR), CAC, and the lecture's ATM **disadvantages**.

Key terms

Term	Meaning in this lecture
ATM cell	5-byte header + 48-byte payload, hardware-switched
UNI / NNI	User–network vs network–network cell formats (GFC vs extra VPI bits)
VPI / VCI	Virtual path and channel IDs used for forwarding (and for VC-stealing attacks)
HEC	8-bit header CRC; framing plus single-bit correct / multi-bit drop
AAL / AAL5	Adaptation layer; AAL5 uses the AUU PT bit for packet boundaries
ABR / CBR / UBR / VBR	ATM bit-rate service classes
VC stealing	Compromised switches swap VPI/VCI to ride another user's QoS
CAC	Connection admission control at setup for guaranteed QoS
Cell tax	10% header overhead vs ~1% on Ethernet
NBMA	Non-broadcast multiple access — ATM has no native broadcast

Exam lines

- ATM's security and performance story starts from **fixed cells and pre-established VCs** on a **SONET-class** fabric — good for **voice/video latency**, awkward for **surges** and **broadcast**.
- Header fields (**VPI/VCI, PT, CLP, HEC**) are both the switching mechanism and the **attack surface** (clear-text parties, CLP drops, false RELEASE).
- Forum-style security is **confidentiality, AAL checksums/sequence numbers, signed signaling, and a full management lifecycle** (log, alarm, audit, recover).
- Unique ATM threats taught here: **VC stealing** between colluding switches and **traffic/covert-channel analysis** despite encryption.
- QoS is a **traffic contract** (PCR/SCR/CLR/CTD/CDV/BT/MCR) plus **CAC**; unspecified QoS exists so the network is not wasted.
- The closer is practical: ATM **lost the LAN** to cheaper, simpler, more interoperable Ethernet despite remaining in some **telco backbones**.

M30: Wireless Networks

Learning focus

- Define a wireless network as radio (OSI physical layer) used to avoid cabling, and classify **WPAN, WLAN, WMAN, WWAN**.
- Describe Bluetooth **IEEE 802.15**: piconet/scatternet, 2.4 GHz ISM FHSS/GFSK, baseband/LMP/L2CAP, and the 72/54-bit frame.
- Explain IEEE 802.11 BSS/ESS, station mobility, **DCF (CSMA/CA)** vs **PCF**, why CSMA/CD fails, 802.11 MAC frame/types, and **hidden vs exposed station** (RTS/CTS).
- Outline WiMAX (IEEE 802.16), WWAN microwave/cellular (cells, BSC/MSC), and satellite networks as taught.

Key terms

Term	Meaning in this lecture
WPAN / WLAN / WMAN / WWAN	Personal, local, metro, and wide-area wireless scopes
Piconet / scatternet	Bluetooth 1-master small net vs overlapping piconets
Parked state	Bluetooth secondary synced but not in the active seven
GFSK / FHSS 1600 hop/s	Bluetooth 2.4 GHz ISM modulation and hopping
BSS / ESS / AP	802.11 building block, extended set, optional access point
DCF / PCF	CSMA/CA distributed access vs AP polling (PIFS < DIFS)
Hidden / exposed station	Collision because sender cannot hear a peer vs false deferral
RTS/CTS	Handshake that advertises duration so hidden nodes defer
WiMAX / IEEE 802.16	Broadband WMAN
MSC	Mobile Switching Center in cellular

Exam lines

- Wireless is classified by **reach**: PAN (Bluetooth 802.15), LAN (Wi-Fi 802.11), MAN (WiMAX 802.16), WAN (microwave, cellular, satellite).
- Bluetooth is a **master-driven hopped piconet** (79×1 MHz, 1600 hop/s, GFSK) with a noisy-radio frame that **votes on a triple header** instead of retransmitting at baseband.
- 802.11 infrastructure is **BSS+AP**; ad hoc is BSS without AP. MAC is **CSMA/CA**, not CD, because of cost, **hidden nodes**, and fading.
- **RTS/CTS** mitigates hidden stations; the **exposed-station** problem is left as a capacity waste the handshake does not fully solve.
- Cellular **low-power cells + MSC** and satellites **cell-ize the planet** so coverage does not require cabling every site.

M31: Wi-Fi Security Protocol

Learning focus

- Distinguish **IEEE 802.11** (PHY/MAC specs) from the **Wi-Fi Alliance** brand, interoperability testing, and required **WPA/WPA2** plus **EAP**.
- Describe **FHSS vs DSSS** (chips), the 802.11 split into **PLCP/PMD**, MAC management, and **DCF (CSMA/CA)** vs **PCF**.
- Tabulate 802.11 / a / b / g / n / ac (and d, e, f, h, i, j, ad, af) with bands, modulation, rates, and ranges the lecture gives.
- Decode the 802.11 **MAC frame** and **Frame Control** bits, including the **WEP** body-encryption flag.

Key terms

Term	Meaning in this lecture
Wi-Fi Alliance	1999 nonprofit that certifies 802.11 interoperability; requires WPA/WPA2 and EAP
FHSS / DSSS / OFDM / HR-DSSS	802.11 PHY families (hops, chips, 54 Mbps+ OFDM, 11/22 Mbps HR-DSSS)
PLCP / PMD	PHY convergence (CCA) vs medium-dependent modulation
CSMA/CA	DCF access: sense idle, send full frame, binary exponential backoff on collision; virtual sensing via MACA
MIMO	Multiple antennas added in 802.11n (54–600 Mbps)
IEEE 802.11i	Security/authentication amendment: RADIUS, Kerberos, 802.1X
WEP	Wired Equivalent Privacy; Frame Control W bit means the body is WEP-encrypted
802.11ad / WiGig	60 GHz PHY, peak 7 Gbps
802.11af / White-Fi	TV white-space WLAN, 54–790 MHz cognitive radio

Exam lines

- **IEEE writes 802.11; the Wi-Fi Alliance owns the logo** and, for certification, insists on **WPA/WPA2 and EAP** — that is this lecture's primary "security protocol" statement.
- 802.11 is **PHY + MAC only**: PLCP/PMD below, DCF/PCF above, with MAC management for **associate, roam, authenticate, encrypt, sleep**.
- Rate/range progression taught: **1–2 Mbps (1997) → a/b (1999) → g (2003) → n MIMO (2009) → ac ~1.3 Gbps (2013)**, plus **ad (60 GHz)** and **af (TV white space)**.
- **802.11i** is the amendment that names **RADIUS, Kerberos, and 802.1X**.
- The MAC header's four addresses and Duration field support **distribution-system forwarding and channel reservation**; the **W bit** flags **WEP** on the body.

M32: Wi-Fi Security Protocol

Learning focus

- State why IEEE 802.11 needed a confidentiality protocol and why WEP failed that role.
- Describe WEP's two authentication methods and the RC4/CRC/IV encryption pipeline.
- List the design flaws, IV weakness, and the mitigations that still left WEP crackable in minutes.
- Explain WPA as draft IEEE 802.11i (TKIP + Michael) and WPA2 as the full 2004 standard (AES).

- Compare WPA and WPA2 on 802.1X/EAP, PSK, mix-mode, cryptography, and processing cost.

Key terms

Term	Meaning in this lecture
WEP	Wired Equivalent Privacy — original 802.11 confidentiality algorithm (1997)
WPA	Wi-Fi Protected Access — 2003 interim / draft 802.11i using TKIP
WPA2	Full IEEE 802.11i-2004 using AES
IEEE 802.11i	WLAN security standard implemented in two steps as WPA then WPA2
Open System Authentication	Join path that does not actually authenticate; encryption keys applied later
Shared Key Authentication	Four-step WEP challenge–response using the same key later used to encrypt
RC4	Stream cipher WEP uses with a 256-entry 8-bit lookup table
IV	Initialization vector concatenated with the WEP key; sent in the clear — the IV weakness
CRC	WEP integrity check on plaintext; too weak to stop alteration
TKIP	Temporal Key Integrity Protocol — per-packet 128-bit keys in WPA
Michael	WPA message integrity check; stronger than CRC, still flawed on short packets
AES	WPA2 block cipher: 128-bit blocks; 128/192/256-bit keys; 9/11/13 rounds as taught
802.1X / EAP	Framework both WPA and WPA2 use for mutual authentication and dynamic keys
PSK	Pre-shared key mode for home / small office
Mix-mode	WPA2 ability to host WPA and WPA2 stations on one network

Exam lines

- Air is shared with hackers; 802.11's first answer was WEP, designed to be **hard** to break and **optional**, not military-grade.
- WEP's shared-key handshake reuses one key for auth and encryption and is **MITM-prone**; RC4 with a **clear IV** and **CRC** leaked keystream.
- Firmware-friendly **WPA/TKIP/Michael** was only a bridge to **802.11i**; Michael still allowed short-packet reinjection.
- **WPA2/AES** is the full 2004 standard: stronger crypto, authentication, key management, replay protection, and integrity — at higher CPU cost.
- Prefer WPA2; WPA is described as hackable, WPA2 as theoretically not. Both still share 802.1X/EAP, PSK, and 802.11a/b/g coverage.

M33: Bluetooth Networks and Security Protocols

Learning focus

- Place Bluetooth as a short-range UHF ISM WPAN technology and recount SIG / IEEE 802.15 origins.
- Draw a piconet and scatternet and state the master/slave communication rule.
- Walk the Bluetooth protocol stack, ACL vs SCO links, L2CAP roles, and the 54-bit header frame.
- Recite SIG versions from 1.0 through 4.2 and the IEEE 802.15.x WPAN family.
- Name Bluetooth's three native security services and the four BR/EDR/HS security modes, including NIST's warning on mode 1.

Key terms

Term	Meaning in this lecture
WPAN	Wireless personal area network; Bluetooth's setting
SIG	Bluetooth Special Interest Group (1998; 1,500-page v1.0 in 1999)
Piconet	Master + up to 7 active slaves within ~10 m
Scatternet	Piconets joined by a bridge node
ACL / SCO	Async connectionless data vs synchronous connection-oriented real-time (no retransmission, FEC)
L2CAP	Segmentation, multiplexing, QoS
RFCOMM	Serial-port emulation
Access code	Identifies the master so overlapping piconets can be distinguished
EDR	Enhanced Data Rate (v2.x): ~3 Mbps nominal, 2.1 Mbps practical, GFSK+PSK
HS	High Speed (v3.0): 24 Mbps over 802.11 after Bluetooth setup
LE / Wibree	Bluetooth Low Energy new stack inside v4.0 "Bluetooth Smart"
SSP	Secure Simple Pairing (v2.1); ECDH link keys in security mode 4
802.15.1–.10	IEEE WPAN family: Bluetooth, coexistence, high/low rate, mesh, BAN, VLC, PAC, KMP, L2 routing
Security modes 1–4	Non-secure; service-level; link-level mandatory crypto; SSP service-level

Exam lines

- Bluetooth is a complete SIG stack on cheap short-range ISM radios; IEEE 802.15 only standardizes PHY/MAC.
- A piconet is strictly **star-shaped**: no slave-to-slave path except through the master (or a scatternet bridge).
- Baseband TDD (625 µs), ACL vs SCO, and L2CAP fragmentation/mux/QoS are the operational core; audio/control can bypass L2CAP.
- Versions add hopping hygiene, EDR, SSP, 802.11 offload, LE, LTE coexistence, and longer PDUs — always backward compatible.
- Native security is **device** auth, confidentiality, and service authorization only. **Never use mode 1** (NIST). Mode 4 modernizes **key agreement** (ECDH) without changing the old auth/encryption algorithms.

M34: Virtual Private Network

Learning focus

- Contrast a leased-line private network with a VPN overlay on a public network.
- Name Access, Intranet, and Extranet VPNs and the four architectural components (client, NAS, VPN server, protocol).
- Reconstruct Layer-2 tunnel setup (PPP, LCP, PAP/CHAP, NCP) and the PPTP, L2F, and L2TP packet stories.
- Distinguish IPsec AH vs ESP and Layer-2 vs Layer-3 tunneling.
- State OpenVPN's SSL/TLS properties and the lecture's protocol preference; explain why firewalls and VPNs are used together.

Key terms

Term	Meaning in this lecture
Private network	Company hosts plus leased telephone lines
VPN	Overlay tunnels + encryption on a public non-secure network
NAS	Network Access Server at the ISP PoP
PAC / PNS	PPTP Access Concentrator / PPTP Network Server
LAC / LNS	L2TP Access Concentrator / L2TP Network Server
Home Gateway	L2F name for the VPN server
PAP / CHAP	PPP authentication used by Layer-2 VPNs
LCP / NCP	PPP link control / network control (IPCP etc.)
GRE	PPTP encapsulation; many firewalls lack it
AH / ESP	IPsec authentication-only vs encrypting payload
OpenVPN	SSL/TLS VPN via OpenSSL; not interoperable with IPsec/L2TP/PPTP
TAP	Virtual Ethernet device for OpenVPN bridging

Exam lines

- VPNs buy private-network **properties** without leased-line **cost**, via tunnels and encryption on the Internet.
- Layer 2 (PPTP/L2F/L2TP) is the dial-up/PPP world and can carry non-IP; IPsec is IP-only, router/firewall oriented, and has **no user authentication**.
- IETF folded PPTP+L2F into **L2TP**; encrypt L2TP with **IPsec** when possible.
- **Avoid PPTP.** Prefer **OpenVPN**; use **L2TP/IPsec** only for quick, non-critical, especially mobile, setups.
- A firewall pair without a tunnel still leaks on the Internet; **firewall + VPN** is stronger than either mechanism alone.

M35: WiMAX Technology and its Security

Learning focus

- Place Bluetooth as a short-range UHF ISM WPAN technology and recount SIG / IEEE 802.15 origins.
- Draw a piconet and scatternet and state the master/slave communication rule.
- Walk the Bluetooth protocol stack, ACL vs SCO links, L2CAP roles, and the 54-bit header frame.
- Recite SIG versions from 1.0 through 4.2 and the IEEE 802.15.x WPAN family.
- Name Bluetooth's three native security services and the four BR/EDR/HS security modes, including the recommendation never to use mode 1.

Key terms

Term	Meaning in this lecture
WPAN	Wireless personal area network; Bluetooth's setting
SIG	Bluetooth Special Interest Group (1998; 1,500-page v1.0 in 1999)
Piconet	Master + up to 7 active slaves within ~10 m
Scatternet	Piconets joined by a bridge node
ACL / SCO	Async connectionless data vs synchronous connection-oriented real-time (no retransmission, FEC)
L2CAP	Segmentation, multiplexing, QoS
RFCOMM	Serial-port emulation
Access code	Identifies the master so overlapping piconets can be distinguished
EDR	Enhanced Data Rate (v2.x): ~3 Mbps nominal, 2.1 Mbps practical, GFSK+PSK
HS	High Speed (v3.0): 24 Mbps over 802.11 after Bluetooth setup
LE / Wibree	Bluetooth Low Energy new stack inside v4.0 "Bluetooth Smart"
SSP	Secure Simple Pairing (v2.1); ECDH link keys in security mode 4
802.15.1–.10	IEEE WPAN family: Bluetooth, coexistence, high/low rate, mesh, BAN, VLC, PAC, KMP, L2 routing
Security modes 1–4	Non-secure; service-level; link-level mandatory crypto; SSP service-level

Exam lines

- Captions for this playlist slot teach **Bluetooth**, not WiMAX: complete SIG stack on cheap short-range ISM radios; IEEE 802.15 only standardizes PHY/MAC.
- A piconet is strictly **star-shaped**: no slave-to-slave path except through the master (or a scatternet bridge).
- Baseband TDD (625 µs), ACL vs SCO, and L2CAP fragmentation/mux/QoS are the operational core; audio/control can bypass L2CAP.
- Versions add hopping hygiene, EDR, SSP, 802.11 offload, LE, LTE coexistence, and longer PDUs — always backward compatible.
- Native security is **device** auth, confidentiality, and service authorization only. **Never use mode 1**. Mode 4 modernizes **key agreement** (ECDH) without changing the old auth/encryption algorithms.

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Volume 04 — M36–M47

Information Security (CEC) · SWAYAM / CEC Information Security **notes-3** — concept exam notes (definitions & what things do). Breadth over lab depth.

M36: Introduction to Mobile IP and Addressing

Learning focus

- Explain why classic IP prefixes break when a host moves, and why DHCP-only renumbering is not enough.
- Define home address vs care-of address and the roles of home agent, foreign agent, and collocated care-of address.
- Describe agent discovery (advertisement vs solicitation) and the agent-advertisement extension fields.
- Walk registration request/reply and the four-path data-transfer process (proxy ARP + tunneling).
- Distinguish double crossing (2X) from triangle (dog-leg) routing and the binding-cache fix plus its stale-cache problem.

Key terms

Term	Meaning in this lecture
Home address	Permanent address tying the host to its home network
Care-of address	Temporary address on the foreign network
Collocated care-of address	Care-of address when the mobile host is its own foreign agent (often via DHCP)
Home agent	Application-layer agent, usually a home-network router, that intercepts and tunnels

Term	Meaning in this lecture
Foreign agent	Application-layer agent on the visited network that delivers tunneled packets
Agent advertisement	ICMP router advertisement plus type-16 extension (care-of list on FA)
Agent solicitation	ICMP request for an agent when advertisements are missing
Proxy ARP	How the home agent impersonates the mobile host on the home LAN
Tunneling	Home agent encapsulates the original IP packet to the foreign agent
Double crossing (2X)	Remote and mobile on the same net; reply path still detours via home
Triangle / dog-leg routing	Remote → home agent → mobile instead of a direct path
Binding / warning	Cache CoA at the correspondent; invalidate it after a move

Exam lines

- IP prefixes assume a **fixed** attachment point; DHCP renumbering breaks configs, DNS, and **in-flight connections**.
- Mobile IP keeps a **permanent home address** plus a **changing care-of address**, with HA/FA (or a collocated CoA) hiding the move.
- Discovery (ICMP advertisement type **16** / solicitation) then registration (type **1**, lifetime zeros = deregister) then four-path transfer with **proxy ARP** and **IP-in-IP**.
- Transparency is complete for the correspondent — and that is why **2X** and **triangle routing** waste path length.
- Binding caches can shortcut the triangle until the host moves; then the home agent must **warn**.

M37: Cloud Computing Architecture

Learning focus

- Define cloud computing versus local storage and versus a “simple remote machine.”
- Recite the essential characteristics without which, the lecture says, a service is not a cloud.
- Contrast private, public, community, and hybrid deployment, including cloud bursting.
- Distinguish SaaS, PaaS, and IaaS (SPI) with the examples given (CRM, App Engine, Azure, AWS).
- Draw the four-layer architecture and state which layers fall under SLAs.

Key terms

Term	Meaning in this lecture
Cloud	Metaphor for Internet-based, elastic, metered remote compute — not a simple remote PC
On-demand self-service	Customer provisions without a human intermediary
Broad network access	Browser, desktop app, or mobile (e.g. iPhone apps) from anywhere
Resource pooling	Multi-tenant dynamic assignment; location opaque except region/country
Rapid elasticity	Expand and release pools as demand changes
Measured service	Metering and audited pay-per-use reports
Private / public / community / hybrid	Four deployment models
Cloud bursting	Hybrid spike handling by drawing public-cloud capacity
SaaS / PaaS / IaaS (SPI)	App / platform / infrastructure service models
Thin vs thick client	Dependent low-power vs adequate local processing
SLA	Contract whose QoS penalties apply to management and hardware layers, not the user's Internet path
OpenStack	Example stack for private-cloud create/deploy/manage

Exam lines

- Cloud is Internet access to pooled, elastic, metered resources — **hire and pay for consumption**, not a new local box.
- **All** listed essential characteristics are required, or it is **not** cloud computing.
- Choose deployment by control and community: private for mission-critical (still disaster/insider-prone), public for scale/cost (Google example), community for shared concerns (banks/traders), hybrid for bursting and mixed sensitivity.
- **SPI:** SaaS uses apps; PaaS deploys onto a vendor-managed stack (App Engine, Azure); IaaS rents raw resources (AWS). Bring-your-own package → **IaaS not PaaS**.
- Four layers from client device to data center; **network is outside SLAs**, **management and hardware are inside**, with hardware affecting penalties most.

M38: Security in Cloud Computing

Learning focus

- Define cloud computing security as procedures, processes, and standards spanning physical and logical issues in SaaS/PaaS/IaaS and public/private/hybrid delivery.
- List data-security and virtual-data-center challenges (multi-tenancy, EU privacy, disk wipe standards, loss of visibility, provider insiders, hypervisor I/O).
- Enumerate cloud **network** challenges from guaranteed bandwidth through location-dependent VLANs.

- Explain how security responsibility shifts as capabilities (and risks) are inherited up the SPI stack.
- Detail SaaS, PaaS, and IaaS (especially private-cloud) security issues named in the lecture.

Key terms

Term	Meaning in this lecture
Cloud computing security	Procedures, processes, and standards for assurance in cloud environments
Trust path	Dependency among parties in Internet applications such as e-commerce
New insider	Provider staff who can see/control customer data without being company employees
Sticky services	Provider lock-in that makes leaving (e.g. incompatible storage APIs) hard
Hypervisor	Translation layer whose I/O and security model must not be assumed safe
Broadcast-domain abstraction	Missing cloud-network feature that forces application rewrites
XML encryption / WS-Security / SSL / TLS	In-transit protections listed for SaaS
Amazon S3	Example where **data at rest are unencrypted by default**
CIA	Confidentiality, integrity, availability — QoS on virtual networks hits **availability**
Active Directory	Example traditional authentication repository that private cloud outgrows

Exam lines

- Cloud security is **physical and logical**, across **SPI** and **public/private/hybrid**, and must answer **where data live and who can see them**.
- Putting data off-premises raises **multi-tenancy, EU privacy, wipe standards, lost SIEM feeds, and provider insiders**.
- Virtual data centers hide load-time failures: hypervisor I/O, **app-unaware** VMs, huge disk images, and **portable images flooding storage networks**.
- Cloud networks struggle with **SLA bandwidth**, **appliance insertion**, **per-switch policy**, **protocol zoo**, **no broadcast domain**, **cloud IPs**, and **VLAN/port location binding**.
- **Risks inherit up the stack**; the lower the provider's security cutoff, the more the customer must implement. SaaS adds browser/transit/locality/sticky/backup issues (encrypt S3 yourself); PaaS adds object isolation and privacy; IaaS/private cloud still has **hypervisor, awkward internal multi-tenancy, IAM, and virtual-network CIA**.

- Security is required at **every interface** among virtualized hardware, infrastructure, platforms, and business apps.

M39: GSM Architecture

Learning focus

- Place GSM as the dominant 2G digital cellular standard (CEPT GSM → ETSI → 3GPP) and list GSM900 / DCS1800 / PCS1900 bands.
- Partition the network into RSS, NSS, and OSS and name the A, Abis, Um, and O interfaces.
- Identify MS hardware vs SIM identifiers (IMEI, IMSI, PIN/PUK, Ki, Kc, TMSI, LAI).
- Explain MSC, HLR, VLR, EIR, AuC, GMSC, and SMS gateways, including HLR/VLR hierarchy.
- Classify GSM teleservices, bearer services (9.6 kbps, HSCSD, GPRS), and supplementary services (CLIP/CLIR, AoC, CUG, …).

Key terms

Term	Meaning in this lecture
GSM	2G digital cellular; ~70% share; 800+ million users in 190+ countries
RSS / NSS / OSS	Radio, switching, and operation subsystems
Um / Abis / A / O	MS–BTS radio; BTS–BSC; BSS–MSC (PCM-30); management (SS7/X.25)
IMEI / IMSI	Equipment identity (in hardware) vs subscriber identity (on SIM)
PIN / PUK / Ki / Kc	Unlock SIM; unblock after 3 PIN fails; auth key; cipher key
TMSI / LAI	Temporary subscriber ID and location-area ID stored while camped
HLR / VLR	Permanent user DB vs per-MSC visitor copy (cuts long-distance signaling)
AuC / EIR	Auth/cipher secrets; IMEI white/black/grey lists (blacklists not shared across operators)
GMSC / SMS-GMSC / SMS-IWMSC	Incoming voice routing; SMS toward MS; SMS from MS
Teleservice / bearer / supplementary	Speech/SMS/fax; 9.6 kbps data path (HSCSD, GPRS); extras like CLIP and CUG
MSISDN / MSRN	Directory number vs routing number obtained from HLR

Exam lines

- GSM is Europe’s answer to fragmented analog 1G: **one digital 2G** under ETSI then **3GPP**, built to **roam** with **ISDN-like voice**, later band-extended as GSM900 / DCS1800 / PCS1900.

- Users only see **MS + BTS antennas**; the real machine is **RSS (Um/Abis) + NSS (A, HLR/VLR/MSC) + OSS (O, TMN)**.
- Identity splits: **IMEI** (stolen phone) vs **IMSI/SIM** (stolen credentials); **VLR copies HLR** so the home database is not hit on every local event.
- Incoming calls go **GMSC → HLR (MSRN) → visited MSC**; SMS has analogous **GMSC/IWMSC** pair.
- Services stack as **teleservices (13 kbps speech, SMS, fax)**, **bearers (9.6 kbps, then HSCSD/GPRS)**, and a long list of **supplementary** controls (CLIP/CLIR, barring, CUGs, AoC).
- EIR blacklists **do not sync between operators**, so a stolen handset may still work on another network.

M40: MANET - I

Learning focus

- Contrast wired networks with wireless networks and name the three wireless types the lecture uses: fixed wireless, wireless with fixed access points, and MANET.
- Define a mobile ad hoc network as an infrastructure-less, IP-based radio network in which nodes act as both hosts and routers.
- List MANET characteristics (autonomous terminals, distributed operation, multi-hop routing, dynamic topology, limited resources and security, scalability, low bandwidth).
- State the open challenges the instructor listed: scalability, routing, QoS, the client–server model shift, security, energy conservation, node cooperation, and interoperation.
- Recall the application families taught: military, home/office/education, VANET, wireless sensor networks, wireless mesh networks, disaster relief, and personal area networks.

Key terms

Term	Meaning in this lecture
MANET	Infrastructure-less, IP-based radio network of mobile nodes that route for one another
Access point	Fixed infrastructure node that mobile hosts associate with in the non-ad-hoc wireless type
Multi-hop	Relaying through intermediate nodes when source and destination are not in direct range
Dynamic topology	Unpredictable, rapid link change because nodes move freely
QoS	Guaranteed performance for a flow: delay, jitter, bandwidth, packet-loss probability
VANET	Vehicular ad hoc network; V2V between vehicles, V2I via roadside units

Term	Meaning in this lecture
WSN	Sensor-node network treated as a low-mobility special case of MANET, with sink nodes
Wireless mesh	Many 802.11a/b/g mesh routers sharing an Internet connection by dynamic hop-by-hop routing
PAN	Person-centred short-range network, including Bluetooth wire-replacement

Exam lines

- Wireless freedom of movement yields three classes; MANET is the infrastructure-less extreme.
- Every MANET node is potentially a host **and** a router; A–C communication may require B.
- Design is dominated by dynamics, scarce energy/CPU/bandwidth, and the absence of a place to hang access control.
- Routing, QoS, naming/addressing without servers, security without a CA, and merging dissimilar networks are still open.
- The same idea covers battlefields, classrooms, vehicles, sensors, city-scale meshes, disasters, and PANs.

M41: MANET – Routing Protocols

Learning focus

- Define what a routing protocol does in a MANET and why metrics and information exchange matter under changing conditions.
- Distinguish **reactive (on-demand)**, **proactive (table-driven)**, and **hybrid** routing, including the bandwidth and latency trade-offs the lecture stated.
- Explain **AODV** (RFC 3561): RREQ / RREP / RERR, expanding-ring TTL, sequence numbers, precursor lists, and the three RERR circumstances.
- Explain **OLSR** as a proactive protocol that uses **multipoint relays (MPRs)**, Hello-based neighbour sensing, MPR flooding, topology-control (TC) messages, and Dijkstra shortest paths.
- Explain **TORA** as a hybrid protocol that builds a **DAG** of **heights** using QUERY, UPDATE, and CLEAR, with route creation, five maintenance cases, and route deletion.

Key terms

Term	Meaning in this lecture
Reactive / on-demand	Build a route only when needed; higher search latency
Proactive / table-driven	Continuously maintain routes to all nodes via periodic control traffic

Term	Meaning in this lecture
Hybrid	Proactive backbone plus reactive flooding for extra demand
AODV	Ad hoc On-Demand Distance Vector (RFC 3561); RREQ, RREP, RERR
Expanding ring	Increase RREQ TTL on retransmission if no reply
Precursor list	Neighbours that likely use this node as next hop; used to send RERR
OLSR	Optimized Link State Routing; proactive, MPR-optimized link state
MPR	Multipoint relay: selected one-hop neighbour that forwards floods to two-hop neighbours
TC message	Topology-control advertisement from nodes with a non-empty MPR selector set
TORA	Temporally-Ordered Routing Algorithm; DAG of ordered heights, loop-free multiple paths
QUERY / UPDATE / CLEAR	TORA control packets for create, maintain, and delete routes

Exam lines

- MANET routing is the algorithm-as-protocol that tracks a moving graph with chosen metrics.
- Reactive protocols save bandwidth but pay **initial delay**; proactive protocols start sessions quickly but **waste bandwidth** and recover slowly from failures; hybrid needs **foreknowledge** of size and demand.
- AODV finds a path when RREQ hits the destination or a node with a valid cache, replies along reverse routes, and uses RERR plus precursor lists when links die.
- OLSR shrinks classical link state by letting only MPRs rebroadcast and by advertising only MPR-selector links; Dijkstra then yields hop-shortest routes.
- TORA orients a DAG with heights so traffic is loop-free; QUERY/UPDATE build routes, five cases repair them, and CLEAR erases a partition.

M42: 3G Network and Security

Learning focus

- Place 3G in the ITU **IMT-2000** story (spectrum, pre-commercial and commercial launches, India's MTNL/BSNL path, and the GPRS → EDGE transition).
- State what 3G is: third-generation mobile telecommunications meeting **IMT-2000**, with at least **200 kbps** and later 3.5G / 3.75G broadband rates.
- List 3G capabilities, applications, and services (mobile Internet, HTML vs plain-text email, SMS/MMS/IM plus presence).
- Sketch 3G architecture: **RAN** (Node B, RNC) and **core** (packet domain SGSN/GGSN, circuit domain MSC, CGF) over **ATM Iu/Iub/Iur** with AAL2 voice and AAL5 data.
- Recall 3G security: CIA, KASUMI vs A5/1, IMSI/TMSI and false-base-station issues, and the **ETSI TS 121 133** threat classes.

Key terms

Term	Meaning in this lecture
IMT-2000	ITU name for 3G technical specifications
EDGE / EGPRS	Enhanced GPRS step before 3G; new 3G frequencies still need new infrastructure
RAN / Node B / RNC	Radio access network; Node B ≈ 2G BTS; RNC ≈ BSC (RRM, handover)
SGSN / GGSN / MSC / CGF	Packet core (as in GPRS), circuit MSC for voice, charging gateway
Iu / Iub / Iur	ATM RAN interfaces; Iu split CS (AAL2 voice) and PS (AAL5 data)
KASUMI	3G block cipher replacing A5/1; still taught as having serious weaknesses
IMSI / TMSI	Subscriber identity; IMSI sent in clear when assigning TMSI
False BS	Rogue base station that camps the user away from legitimate paging
ETSI TS 121 133	Specification used to classify 3G confidentiality, integrity, DoS, repudiation, and service-access threats

Exam lines

- 3G is IMT-2000: ≥200 kbps (later several Mbit/s), new spectrum and infrastructure, not just a software bump from EDGE.
- The architecture splits RAN (Node B + RNC) from a dual CS/PS core, with ATM AAL2/AAL5 on Iu.
- User-facing 3G is mobile Internet, email (spam/virus fears), SMS/MMS/IM+presence, and location services — with sharp regional MMS uptake differences.
- Mutual authentication is an improvement over earlier generations, but IMSI exposure, false base stations, optional encryption, and KASUMI weaknesses remain.
- ETSI TS 121 133 organises 3G threats around CIA plus repudiation and unauthorized service access.

M43: 4G LTE

Learning focus

- Define **LTE (Long Term Evolution)** as the 3GPP 4G standard on the GSM → UMTS path, aimed at ~10× 3G speeds via new DSP and modulation, and an **IP-based** architecture with lower latency.
- Place LTE among competing 4G candidates (**UMB**, **IEEE 802.16 / WiMAX**, **802.16m / WiMAX Advanced**) and **LTE Advanced (Release 10)**.
- Separate marketing "4G" (**HSPA**) from **4G LTE**, and state OFDM, peak-rate, and VoLTE/RCS benefits as taught.
- List LTE technical objectives (throughput, latency, simplified **E-UTRAN**, mobility) and the two resources required: a capable **network** and a capable **device**.
- Recall the lecture's LTE security framing: confidentiality and integrity, plus research goals for comparing LTE cryptographic algorithms.

Key terms

Term	Meaning in this lecture
LTE	Long Term Evolution; 3GPP 4G on the GSM/UMTS path
LTE Advanced	3GPP **Release 10**, the release taught as qualifying as 4G
E-UTRAN	Simplified LTE radio architecture: no RNC, no CS domain, no DCH
HSPA	High Speed Packet Access — "4G" branding without LTE; faster 3G GSM, slower than LTE
OFDM	Orthogonal frequency-division multiplexing; why 4G is faster
VoLTE	Voice over LTE: IP voice on the LTE data network
RCS	Rich Communication Services launched from the native dialer
WiMAX / 802.16m	Competing 4G path (WirelessMAN-Advanced / WiMAX 2)
UMB	Ultra Mobile Broadband, another competing 4G standard

Exam lines

- LTE is 3GPP's IP redesign of the GSM/UMTS family: DSP/OFDM speed, simpler E-UTRAN, sub-5 ms user-plane latency in the ideal case.
- Marketing "4G" may be HSPA; **4G LTE** is taught as the pure 4G form, still needing both network and device — and the right spectrum.
- VoLTE is not only HD codecs (8 vs 13 kbps) but RCS, faster setup, better battery than Skype-class OTT, and a path to VoWiFi and cross-carrier handset interoperability.
- Security is framed as CIA plus a research programme over LTE cryptographic algorithms, not a finished protocol walkthrough in this lecture.

M44: 5G

Learning focus

- Define **5G** as fifth-generation mobile/wireless broadband as the lecture presented it, including the **IEEE 802.11ac** claim and the generation path from 1G through 4G LTE.
- State **NGMN Alliance** rate, connection-count, spectral-efficiency, coverage, signalling, and **~1 ms latency** requirements.
- Describe the taught **all-IP 5G architecture**: user terminal plus multiple simultaneous **radio access technologies (RATs)**, software-defined radios, and terminal-side choice via middleware.
- Contrast 5G with 4G on speed, capacity (~1000×), latency, millimetre-wave vs existing bands, MIMO, and small cells.
- List benefits, device/network requirements, applications (IPv6, global coverage), disadvantages, and the "nanocore / 6G" closing remarks — all as taught.

Key terms

Term	Meaning in this lecture
5G	Fifth-generation mobile/wireless; lecture also ties it to IEEE 802.11ac and a 5 GHz signal
NGMN Alliance	Source of 5G rate, massive-IoT connection, spectral, coverage, and ~1 ms latency requirements
RAT	Radio access technology; each appears as an IP link; multiple active interfaces at once
MIMO	Multiple-input multiple-output, to boost shorter-range higher-frequency 5G signals
Millimetre wave	Very high frequencies (taught as 3–300 GHz): fast, little broadcast coexistence, poor through walls — hence small cells
LTE vs WiMAX	Competing 4G; LTE became the consistent standard 5G builds on
IPv6	Addressing for mobile IP tied to network and geography in the 5G application vision
Nanocore	Speculative future 5G enhancement combined with AI

Exam lines

- 5G is taught as all-IP, multi-RAT, terminal-centric: the phone keeps several radios up and chooses a provider; networks handle mobility; vertical handovers are to be avoided.
- NGMN targets tens of thousands of users, massive sensor connections, better spectral/signalling efficiency, and about **1 ms** latency.
- Speed/capacity claims in the lecture range from 500+ Mbps typical, 1–10 Gbit/s peaks, ~1000× 4G capacity, and Huawei's "100× fastest LTE" / sub-1 ms latency — with the caveat that **no universal standard existed yet**.
- Higher bands (up to 6 GHz and mmWave) buy rate but lose range, so MIMO and dense small cells (including bounce-off-surfaces coverage) are part of the story.
- Benefits (IoT, AR, film-in-a-second, bigger data plans) sit beside unsolved security/privacy, device replacement cost, and unfinished research.

M45: VoIP Protocols

Learning focus

- Define **VoIP** as packetized digital voice/multimedia over IP instead of circuit-switched TDM, and explain why IP alone is a poor realtime transport.
- Place VoIP against the **OSI** layered model and name the three signalling/control protocols taught: **ITU-T H.323**, **IETF SIP**, and **MGCP**.
- Trace sender-to-receiver processing: codecs **G.711, G.729, G.723**, RTP/UDP/IP headers, playout buffer, and SIP/H.323 for call setup/teardown.
- List **QoS** factors: delay (source, receiver, network), jitter, packet loss, echo, and throughput.
- Recall VoIP **threat classes** (DoS, sniffing, eavesdropping, spoofing, toll fraud, SPIT) and the **defences and deployment options** the instructor listed — not attack recipes.

Key terms

Term	Meaning in this lecture
VoIP	Packet voice/multimedia over IP instead of circuit TDM
H.323	ITU-T umbrella (H.225, H.235, H.245, H.450) with RTP/RTCP over UDP and TLS setup
SIP	IETF RFC 3261 text request/response signalling; needs TLS/IPsec because proxies must see headers
MGCP	RFC 3435 gateway control; IPsec AH/ESP between media gateway and MGC
G.711 / G.729 / G.723	Example voice codecs on the send path
Playout buffer	Receiver jitter buffer; late packets discarded
SPIT	Spam over Internet Telephony — real-time voice spam
Toll fraud	Unauthorized use of paid voice service, often via signalling or configuration abuse
PSTN	Public Switched Telephone Network; still required for number translation and non-VoIP users

Exam lines

- VoIP wins on TDM cost and voice–data integration, but IP is connectionless and offers no realtime guarantees — hence codecs, RTP, jitter buffers, and signalling (H.323, SIP, MGCP).
- H.323 is a full ITU stack with H.235 authentication variants and Diffie–Hellman; SIP is lightweight and **cannot hide routing headers from proxies**, so TLS/IPsec carry the security burden; MGCP assumes IPsec to the PSTN gateway.
- QoS is delay, jitter, loss, echo (electrical vs acoustic), and throughput — not a single number.
- Threats taught are DoS, sniffing, eavesdropping, caller-ID spoofing/phishing, toll fraud, and SPIT; responses are authentication, encryption, filtering/blacklists, firewalls, physical/port security, and user caution — not protocol exploits.
- Deployment spans USB adapters, softphones, dedicated phones/routers, and wireless LAN, with scalability, PSTN interworking, and incomplete encryption as the practical limits.

M46: Introduction to DDoS

Learning focus

- Distinguish **DoS** from **DDoS** and place both as attacks on **availability** in the CIA triad.
- Explain why Internet growth and a security-light original design made flooding and related availability attacks severe.
- State attacker **motives** and organisational **consequences** as listed in the lecture.

- Name the **protocol/architecture weaknesses** the instructor blamed (stateless forwarding, lack of authentication / IP spoofing, predictable TCP handshake, mimicry of legitimate users).
- Classify DDoS **as taught**: physical (layer 1), volumetric (layers 2–4), and application-layer (layers 5–7) — class names only, not operational recipes.

Key terms

Term	Meaning in this lecture
DoS	Denial of service: prevent legitimate users from using a service (availability)
DDoS	DoS whose traffic comes from **many sources**, amplifying impact and complicating defence
Bot / botnet	Compromised machines (masters, handlers, zombies) used as a distributed attack network
IP spoofing	Forged source addresses; hides origin and defeats simple identity-by-IP defences
SYN flooding	Volumetric class that leaves TCP connections **half-open** until timeout
Volumetric attack	Layers 2–4 congestion/bandwidth exhaustion
Application-layer DDoS	Layers 5–7: low network volume, high application/request load
Ping of Death	Named vulnerability class: oversized ICMP mishandled by some OSes; patchable
CIA	Confidentiality, integrity, availability — DDoS targets availability

Exam lines

- Availability was historically under-weighted; flooding proved it can bankrupt, embarrass, or politically coerce a victim.
- DDoS is harder than DoS because of **scale**, **legitimate-looking traffic**, and **spoofed identity**.
- The Internet's **stateless forwarding**, **weak authentication**, **predictable TCP handshake**, and **easy mimicry of users** are the taught root causes — not a single bug.
- Classification is **physical vs volumetric vs application-layer**; named floods (MAC, ICMP, SYN, UDP, HTTP GET/POST) are **types**, not procedures.
- Patching stops some vulnerability DoS; **public-Internet flooding has no easy preventative**, and commercial tools have not closed the problem.

M47: Defence to DDoS Attacks

Learning focus

- Explain why **DDoS flooding** is hard to defend: easy launch, traffic similarity, IP spoofing, huge volume, many zombies, and Internet **hub-and-spoke** chokepoints.

- Separate **technical** vs **social** defence challenges (distributed response, missing incident data, no benchmarks, large-scale testing; deployment economics).
- List **goals of an ideal defence**: effectiveness, completeness, minimum collateral damage, low false positives, low cost.
- Name the four **defence modules**: prevention, detection, source identification, and reaction — as strategies, not attack methods.
- Recall the instructor's **hardening tips** (filters, patches, unused services, quotas, baselines, physical security, integrity tools, redundancy, backups, passwords).

Key terms

Term	Meaning in this lecture
IP spoofing (defence view)	Makes new legitimate clients indistinguishable from attack initiators
Collateral damage	Legitimate traffic dropped or delayed by a defence
False positive	Defence treats a non-attack (e.g. flash crowd) as DDoS
Flash event	Sudden legitimate load that must be distinguished from DoS
SYN cookies	Example partial preventive for SYN floods; not complete by itself
Anomaly-based detection	Preferred DDoS detector; signatures change too often
Source identification	Finding origins even when addresses are forged
PlanetLab / Emulab	Named research testbeds; small-scale tests are not credible
Tripwire	Example integrity tool for detecting configuration-file changes

Exam lines

- Flooding looks like a "perfect crime": cheap volume, spoofed identity, too many bots to punish, and hubs that can congest the whole Internet.
- A box next to the victim cannot keep up; real defence must be **distributed**, but **economics and reputation-secrecy** block the data, benchmarks, and deployment that research needs.
- Ideal properties are **actually stopping attacks**, covering **many attack styles**, **not punishing legitimate users** (including flash crowds), **few false alarms**, and **cost in line with benefit**.
- The operational pipeline is **prevent (especially anti-spoof near sources) → anomaly-detect → identify sources → react without collateral damage**.
- Everyday hygiene — filters, SYN-flood patches, turning off spare services, quotas, baselines, physical security, Tripwire, redundancy, backups, strong admin passwords — **reduces exposure**; it is **not** a complete DDoS solution, which the instructor says the market has not yet delivered.
